

Global Supply Chains and Regional Shocks*

Nadim Elayan-Balagué[†]

José Ramón Morán[‡]

Banco Central de Chile

Banco de México

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Abstract

Firms assembling complementary inputs face a trade-off in where to source them. Distant suppliers specialize in more distinct products, but their supply disruptions are less synchronized, and when inputs are complements, failures arriving separately cost more than failures arriving together. Using Mexican customs data, we treat the COVID-19 pandemic as a natural experiment in staggered regional disruption, comparing inputs hit at different times within the same firm. Firms respond along two opposing margins: sourcing inputs from a more common set of regions, while adding backup suppliers for each. We rationalize this in a quantitative model of global input sourcing where cost draws are correlated within regions and delivery failures share regional factors. Complementarity matters only when exposure is uneven across inputs, and comparative advantage generates that unevenness. We quantify the model to ask how much a change in perceived risk explains the post-2020 nearshoring of Mexican supply chains.

JEL Codes: F12, F13, F15, F60

Keywords: supply chains, nearshoring, supply disruptions, global sourcing, correlated comparative advantage, correlated risk

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[†]Research Department, Agustinas 1180, 8340454 Santiago, Región Metropolitana, Chile. Contact at <https://nadimelayan.github.io>, or at nelayan@bcentral.cl.

[‡]Research Department, Avenida 5 de Mayo 18, Colonia Centro, CP 06000, Ciudad de México, México. Contact at <https://joseramonmoran.github.io>, or at jmoran@banxico.org.mx.

1 Introduction

Globalization has ~~witnessed a notable acceleration in recent decades~~ grown rapidly over the last few decades, particularly in the integration of productive processes. Capitalizing on countries' comparative advantage patterns, production has progressively fragmented across the globe. For instance, steel production may occur in China, followed by transformation into an engine in the US, assembly into a car in Mexico, and eventual export back to the US.

In Figure 1, we present the trajectory of total world trade in trillion US dollars from 1988 to 2019 in the left panel, alongside the evolution of world trade in intermediates in the right panel. The surge is remarkable: over 30 years, global trade volume has surged from 1.46 trillion US dollars, accounting for 7.5% of world GDP, to 40 trillion US dollars, constituting 45% of world GDP. The growth in intermediate trade mirrors this trend, climbing from 0.37 trillion US dollars and 1.9% of world GDP in 1988 to 8 trillion US dollars and 9% of world GDP in 2019.

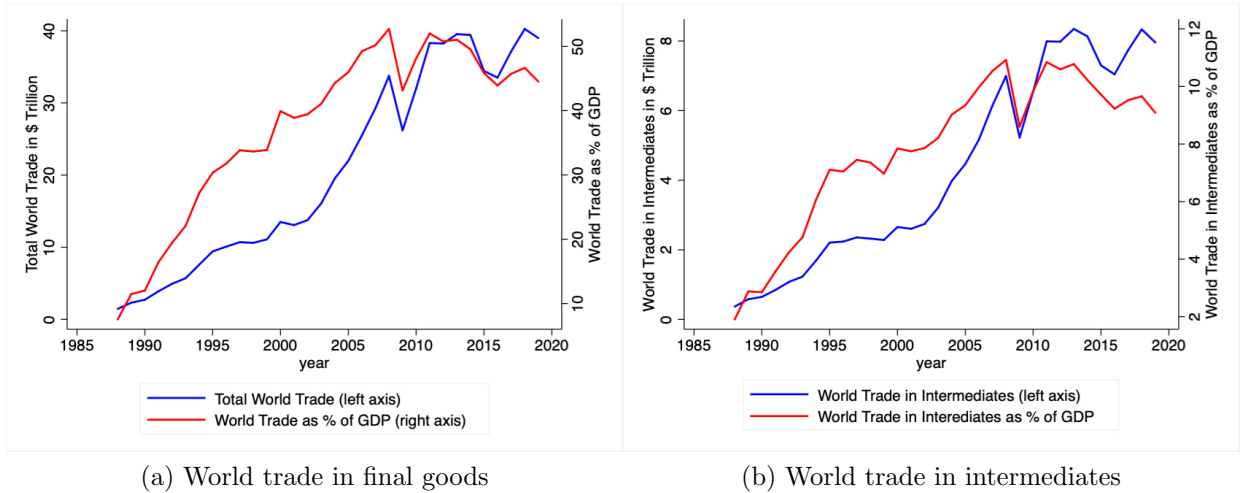


Figure 1: Evolution of world trade in final goods and intermediates.

Global Supply Chains emerge because firms can improve efficiency by locating different stages of production in countries with comparative advantages and lower costs. However, greater fragmentation also increases the risk of disruptions, as production depends on the timely delivery of interconnected inputs from multiple locations. These costs can be particularly severe when inputs are essential or complementary, since a disruption in one stage may halt the entire production process. As a result, firms face a tradeoff between the efficiency gains from more dispersed production and the increased vulnerability that such dispersion creates.

In theory, the reason for why these production chains emerge stems from technological advancements and reductions in trade and transportation costs, enabling the exploitation of global comparative advantages. Building upon our previous illustration, the decision to assemble an engine in Mexico rather than the US may stem from cost advantages, prompting US firms to capitalize on this opportunity. In essence, the benefits of Global Value Chains (GVCs), or the integration of productive processes, lie in firms' ability to enhance production efficiency. However, what about the associated costs? We contend that heightened fragmentation of the production process may elevate chain vulnerability, thereby potentially disrupting the entire production cycle. This vulnerability may materialize in scenarios where a chain relies on essential and interdependent inputs sourced from multiple locations, with the final product contingent upon the seamless provision of these inputs from each location. In this paper we focus on how these two opposing effects—increased efficiency versus increased risk of failure in the production process—shape firms' decisions on how dispersed their GVCs are.

Consider the following extreme scenarios in order to better illustrate our main tradeoff. Consider a scenario with 10 potential production locations, each with an idiosyncratic probability of failing to deliver their input of 10%. In this setup, if any location fails to deliver, the entire chain fails to produce the final good. The likelihood of a production failure hinges crucially on the correlation of these failure probabilities across locations. For instance, in the case of perfectly positive correlation—where shocks are synchronously aligned—the chain would fail only once every 10 periods. Conversely, if failure probabilities are completely independent, the chain would fail, on average, 6.5 periods out of 10. Finally, under a scenario of perfectly negative correlation, the chain would fail in every period.

We propose a mean-variance trade-off framework for supply chain formation, with geographical distance between countries serving as a key determinant. On one hand, firms are incentivized to choose supply chains with countries specializing in different segments of the production process to maximize comparative advantage gains. Our aggregate analysis shows that greater geographical distance between countries is associated with more divergent comparative advantage patterns, thus encouraging firms to source from countries specializing in disparate industries, likely situated further apart. On the other hand, firms also have incentives to form supply chains with countries experiencing more synchronous shocks, particularly in cases involving complementary or essential inputs. We find that countries situated closer to each other tend to have more synchronized business cycles and shocks. Consequently, firms seek to minimize failure rates by sourcing from geographically proximate countries.

We use customs data from Mexico at the exports and imports transaction level to show that firms with more geographically dispersed supply chains tend to exhibit higher export

volumes. However, these firms also experience greater volatility and disruptions in their chains, on average. To quantify the firm-level dispersion of supply chains, we introduce a novel measure based on the geographical distance between all locations from which a Mexican firm sources its inputs.

We provide empirical support for our proposed trade-off at both the aggregate country level and the disaggregated firm level. At the country level, we observe that pairs of countries that are more distant from each other tend to specialize in more diverse industries, whereas pairs of countries that are closer geographically tend to exhibit more similar comparative advantage patterns. Additionally, we find that more distant country pairs experience less correlated (or more negatively correlated) economic shocks, whereas geographically closer country pairs tend to have more synchronized business cycles. At the firm level, we find that firms with more dispersed global value chains (GVCs) tend to achieve higher average export volumes. However, they also exhibit higher levels of volatility. Notably, we observe this trade-off primarily in more complex sectors, where production is presumed to rely more heavily on complementary inputs. This suggests that the trade-off is particularly prominent in environments where GVC locations supply complementary goods to a final goods producer, magnifying the impact of chain disruptions.

We develop a structural model of global input sourcing that incorporates our main mechanism. Building upon the framework of Antràs et al. (2017), our model introduces two key elements. Firstly, we incorporate a multivariate Fréchet distribution with constant elasticity of substitution (CES) region nests of countries, which underlies the comparative advantage in input production. This feature encourages firms to source intermediates from countries that are geographically distant from each other, enhancing the exploitation of comparative advantage gains. Secondly, we consider that firms are exposed to regional shocks in the arrival of the inputs needed to produce their output. Given that shock synchronization is negatively correlated with geographical distance, firms are further motivated to form GVCs with less distant locations to minimize the risk of failures in delivery.

We show that in industries heavily reliant on complementary or essential inputs for production, a tradeoff arises in selecting sourcing locations. Highly dispersed GVCs tend to better exploit comparative advantages, resulting in higher average exports. However, these GVCs also tend to involve countries with less positively synchronized business cycles, leading to more frequent disruptions in the supply chain. These findings are particularly relevant in the aftermath of significant shocks like COVID-19, during which policymakers have intensely debated the merits of renationalizing production. We suggest that an alternative response to mitigate fears of supply chain disruptions could be the development of less global and more regional GVCs.

This paper contributes to the general literature regarding GVCs. We use Mexican customs data for our empirical strategy, as do Antràs (2020), Antràs and de Gortari (2020), de Gortari (2019; 2020), Wang et al. (2017). The focus of these papers is on the length, upstreamness, value added, and the impact of trade agreements on global value chains; we instead focus on a novel mean-variance trade-off that firms face when deciding where to source their inputs from. We also document novel empirical facts about recent trends in GVC size, dispersion, and risk.

Our study contributes to two main branches of the literature. Firstly, we address a very relevant and prevailing topic regarding the status of GVCs in the aftermath of the COVID-19 crisis. By introducing a novel perspective and identifying an empirically unexplored trade-off, we offer an original assessment of the future trajectory of supply chain processes between nations. Secondly, we provide empirical evidence of a theoretical trade-off, as posited by Costinot (2009), between comparative advantage gains and the risk of failure arising from production fragmentation. Expanding upon this idea, we introduce the concept of shock synchronization between countries, thereby uncovering new implications for GVC formation.

This paper aims to answer a policy relevant question: how does the presence of regional shocks and uncertainty affect the formation of GVCs? Miroudot (2020) argues that during the COVID-19 pandemic industries in Thailand that suffered larger economic losses were not the ones with more complex or fragmented production processes and therefore the way we currently understand GVCs should not change. Evenett (2020) suggests that GVCs tend to be in danger during recessions as societies turn to more protectionist measures promising more robustness and less volatility. Bonadio et al. (2021) find that although the evident disruption of global supply chains magnified the last economic crisis re-nationalizing production would have not helped, in fact, they show that it would have made countries more dependent on domestic inputs that were also subject to lockdowns. Eppinger et al. (2020) and Hayakawa and Mukunoki (2021) also find that although the impact of GVC on the economy has been substantial during the global pandemic, re-nationalization would have brought even larger losses. We contribute to this discussion by empirically showing that firms can form more resilient GVCs without necessarily altering the size of them through increasing its shock synchronization. We also allow firms to choose endogenously the size and complexity of their GVCs so that the choice is no longer a binary between complete autarky or a fully globalized process.

Second, we provide empirical evidence of Costinot’s (2009) trade-off between comparative advantage gains and risk of failure derived from the size of a production process, but in terms of GVC formation between countries. Sequential production processes with essential stages carried out in different locations à la Costinot et al. (2013) provide a better intuition for the

main trade-off of our paper. These authors treat size and complexity of GVCs as exogenous whereas in our paper firms will decide endogenously according to comparative advantage and risk forces. Moreover, the risk of failure in a GVC will also come from a novel source, shock synchronization among GVC members.

This paper is structured as follows. Section ?? introduces a simple model that underpins the core trade-off examined in this study. Section 2 outlines the empirical methodology and data sources employed to demonstrate this trade-off. Specifically, we use Mexican customs data, which provides detailed firm-level information on the sourcing of foreign inputs. Additionally, we draw on data from UN Comtrade and the World Input-Output Database (WIOD) to highlight aggregate trends regarding the geographical distance between country pairs and its impact on comparative advantage patterns and business cycle synchronization. Section ?? presents our primary model, built upon the framework proposed by Antràs et al. (2017). Section 5 concludes.

2 Empirical evidence

This section has three objectives. First, we document at the country level that the two forces underlying our mechanism—comparative advantage and shock synchronization—are both systematically related to geographic distance, and in opposite directions. Second, we construct firm-level measures of the structure of a supply chain and establish a set of new facts about how Mexican firms’ sourcing networks evolved between 2014 and 2026. Third, and centrally, we use the staggered regional incidence of the COVID-19 pandemic to identify how firms restructure their sourcing networks in response to a shock to regional supply risk.

Our identification strategy departs from the approach most common in this literature, which compares more- and less-exposed firms before and after a disruption. Because the pandemic coincided with a collapse in external demand, with domestic lockdowns, and with the entry into force of the USMCA, firm-level comparisons of this kind are difficult to interpret. We instead exploit variation *within* firms, *across* the distinct products they import. The pandemic arrived at different times and with different severity in different parts of the world; a firm importing one component from Guangdong and another from Bavaria was disrupted in different months for different inputs. Comparing the subsequent evolution of sourcing for the two inputs, within the same firm and month, holds fixed every firm-level confound by construction.

2.1 Data

Customs data: Our primary source is the universe of Mexican customs transactions from **January 2014 to May 2026**, accessed through the Econlab at Banco de México. For each transaction we observe a firm identifier, the product at the HS 6-digit level, the value of the transaction, the country of origin or destination, and a classification of the good as intermediate or final. We restrict attention to imports of intermediate goods, which reveal firms’ sourcing strategies, and use the same firms’ exports both to assign sectors and as an outcome variable.

We aggregate transactions to the firm $\times$ product $\times$ origin $\times$ month level, denoted M_{fijt} . Our main estimation sample runs from **January 2014** to December 2023. Section 2.4.3 explains this choice, and Section 2.2.8 reports results through May 2026 as a persistence check.

Sector assignment and processing status: We assign each firm to a sector using the modal HS chapter of its exports over the pre-pandemic period, holding the assignment fixed thereafter so that sector is not itself an outcome. A substantial share of Mexican customs flows corresponds to temporary imports for subsequent re-export under the IMMEX program. Because the sourcing decisions of IMMEX establishments may be made by a foreign parent rather than by the Mexican entity, we flag IMMEX status and report all results separately for IMMEX and non-IMMEX firms.

Trade flows: We use the BACI dataset from CEPII, which reports bilateral trade at the HS 6-digit level, to construct revealed comparative advantage indices and to build a leave-one-out measure of realized supply disruptions described in Section 2.4.1.

Gravity: We use the CEPII Gravity database for bilateral distance. We use population-weighted bilateral distance, which is materially more accurate for large and geographically dispersed sourcing countries such as the United States or China.

~~**Business cycle synchronization:** We use World Bank data in current USD to extract the cyclical component of output for each country following Hodrick and Prescott (1997). Because the properties of the Hodrick-Prescott filter on short annual series are known to be poor (Hamilton 2018), we report as a robustness check an alternative constructed from monthly industrial production where available, and we treat the COVID stringency correlations described below as our preferred high-frequency proxy for shock synchronization.~~

Pandemic severity: We use the Stringency Index from the Oxford COVID-19 Government Response Tracker, available at the country-day level, which we aggregate to country-month. The staggered international incidence of the index is the source of identifying variation exploited in Section 2.4.3.

Sectoral technology parameters: Our model in Section 3 will make clear that sectoral heterogeneity operates through two distinct parameters: the elasticity of substitution across input types, ζ_s , and the effective number of input types, N_s . We measure ζ_s using the sector-specific estimates in Gosselin (2025), mapped from NAICS to HS, and we measure N_s as the effective number of distinct intermediate inputs in the **INEGI Mexican input-output matrix**, that is, the inverse Herfindahl of a sector’s intermediate cost shares.

2.2 Measuring the structure of a supply chain

Our empirical exercise requires firm-level measures of how large, how geographically organized, how efficient and how risky a sourcing network is. This subsection develops them.

2.2.1 Three distinct objects

Discussions of “supply chain complexity” routinely conflate three things that our model will keep separate and that have different empirical counterparts.

1. Extent: How many distinct inputs does the firm require, and from how many locations does it source them? In the model this will correspond to N_s , the number of essential input types, and to the cardinality of the sourcing set $\mathcal{J}$.
2. Structure: How is sourcing organized across those inputs and locations? Is each input single-sourced or backed up? Are different inputs drawn from the same regions or from different ones? In the model this will correspond to the sourcing shares $\{\pi_j^m\}$ and, in particular, to exposure dispersion Δ .
3. Complementarity: Conditional on extent and structure, how costly is the failure of a given input? This is a property of technology, not of the network, and corresponds to ζ_s .

The distinction matters because these objects have different, and sometimes opposing, implications. A firm importing two hundred distinct products from two hundred countries, one each, and a firm importing one product from two hundred countries, are identical in (A) and at opposite extremes of (B); the first is fragile and the second is not.

Throughout, let M_{fpjt} denote the value of firm f 's imports of product p from origin j in month t . Let s_{fjt} be the share of origin j in firm f 's total intermediate imports, s_{fjt}^p the share of j within product p , and b_{fpt} the share of product p in the firm's total intermediate imports. Regions are indexed by k and defined in [Appendix ??](#).

2.2.2 Sourcing breadth and input variety

We measure sourcing breadth by the *effective number of source countries*,

$$n_{ft} \equiv \left(\sum_j s_{fjt}^2 \right)^{-1} \quad (2.1)$$

and input variety by the *effective number of distinct inputs*,

$$v_{ft} \equiv \left(\sum_p b_{fpt}^2 \right)^{-1} \quad (2.2)$$

Both are inverse Herfindahl indices, and both are preferred to raw counts for the same reason: raw counts are dominated by a long tail of economically trivial origins and products, and are mechanically increasing in firm size. A firm that imports 99% of its inputs from the United States and makes one token shipment from each of forty other countries has a raw count of forty-one and an effective count barely above one. It is the latter that corresponds to the object in our model.

v_{ft} is the firm-level empirical counterpart of N_s . For example, an assembler of electronic equipment imports hundreds of distinct inputs, each of which must arrive for output to be produced, whereas a sugar refiner imports a handful. We validate v_{ft} against the sector-level N_s constructed from [the INEGI input-output matrix](#) and report the correlation in [Appendix ??](#). Both (2.1) and (2.2) remain positively correlated with firm size, because larger firms buy more of everything. Since our mechanism concerns the structure of sourcing rather than its scale, we residualize both measures on log total imports and log employment within sector $\times$ year cells, and report results using both raw and residualized versions. Without this step, a finding that “complex” supply chains are more volatile is difficult to distinguish from a finding that large firms are more volatile.

Figure 2 reports the effective number of source countries n_{ft} and the effective number of distinct inputs v_{ft} , averaged across firms in each year. The first fact establishes a level rather than a trend: the sourcing network of the average Mexican importer is small. The unweighted mean of n_{ft} sits near 1.6 throughout the sample and that of v_{ft} near 3, both far below the corresponding raw counts in Table 1. The typical importer buys a handful of distinct products from what is effectively one dominant origin with a fringe of secondary

ones. The second fact is the gap between the unweighted and the value-weighted series. Weighting firms by total intermediate imports raises average sourcing breadth from about 1.6 to between 2.4 and 2.9 effective origins, and average input variety from about 3 to between 6 and 7.5 effective products. Aggregate exposure to foreign supply is therefore concentrated in a small number of large importers whose chains are roughly twice as broad as the median firm’s. A claim about “Mexican supply chains” is consequently a claim about one of two quite different populations, and we report both weightings throughout for that reason.

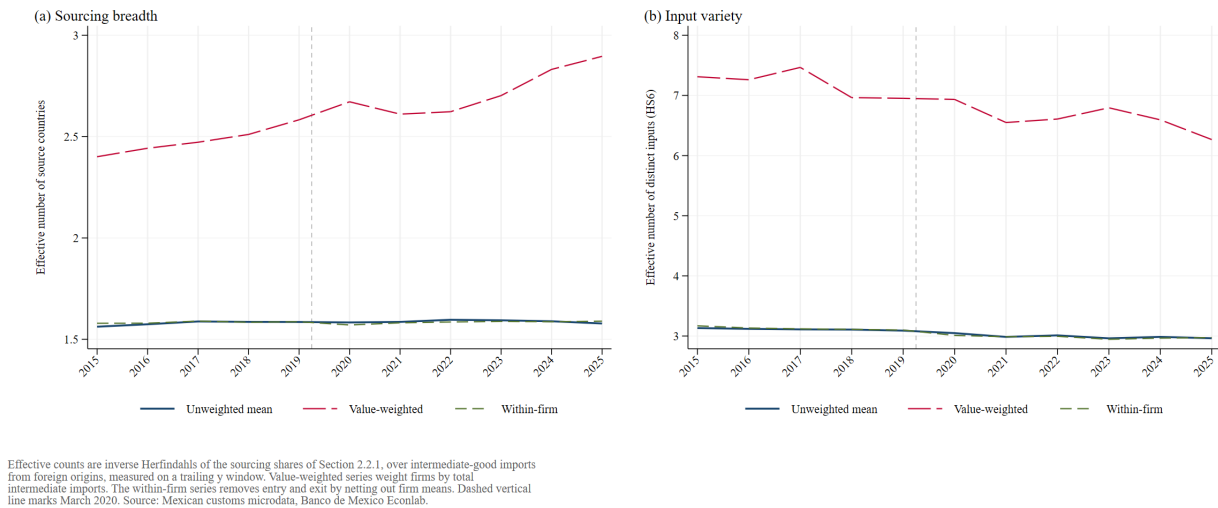


Figure 2: Effective number of source countries and distinct inputs.

The two weightings also move differently over time, and the divergence is itself informative. The unweighted mean of n_{ft} is close to flat: it rises by roughly two percent between 2015 and 2017 and thereafter varies within a band of similar width. The value-weighted mean rises by about twenty percent over the full sample, with a visible interruption in 2020–21 and an acceleration after 2022. Input variety moves in the opposite direction, and does so for both populations: v_{ft} declines from the start of the sample, with the decline concentrated among large importers, whose value-weighted average falls from roughly 7.5 in 2017 to roughly 6.2 in 2025. Read together, the panels say that large Mexican importers have come to source a narrowing set of inputs from a widening set of countries. That combination is suggestive of the redundancy margin our model emphasises—more origins per input rather than more inputs—but the aggregate series cannot separate the two, and the decisive evidence is the within-input decomposition of Section 2.2.6.

Two threats to reading these series as firm behaviour can be assessed directly, and they receive different answers. The first is composition. Figure 3 shows that the number of

firms importing intermediates falls by about five percent in 2020, from roughly 52,900 to 50,000, and does not regain its pre-pandemic level within the sample; since exiting firms are unlikely to be a random draw, the cross-firm mean could move without any firm changing its behaviour. The within-firm series in Figure 2, which nets out firm means and is therefore free of entry and exit, is almost indistinguishable from the unweighted mean. Composition is not what drives the unweighted series. The second threat is scale, and here the answer is different. Figure 4 plots the raw series against the same series residualized on log total intermediate imports within sector. Before 2020 the two move together. After 2020 they separate: raw sourcing breadth rises through 2022, while breadth net of firm size declines continuously and ends some three percent below its 2019 level. The apparent post-pandemic recovery in the number of origins the average firm uses is thus a scale effect—the firms that survived and grew bought more of everything, including from more places—and conditional on size, chains narrowed rather than widened. The same exercise applied to v_{ft} yields a decline that is steeper net of size than in the raw data.

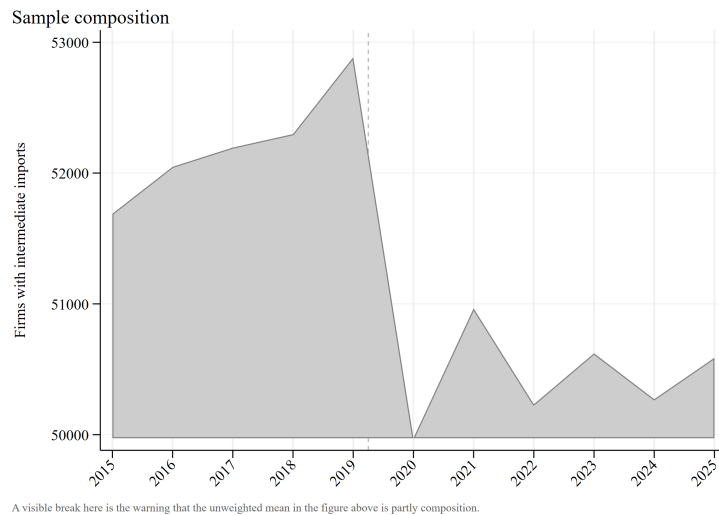
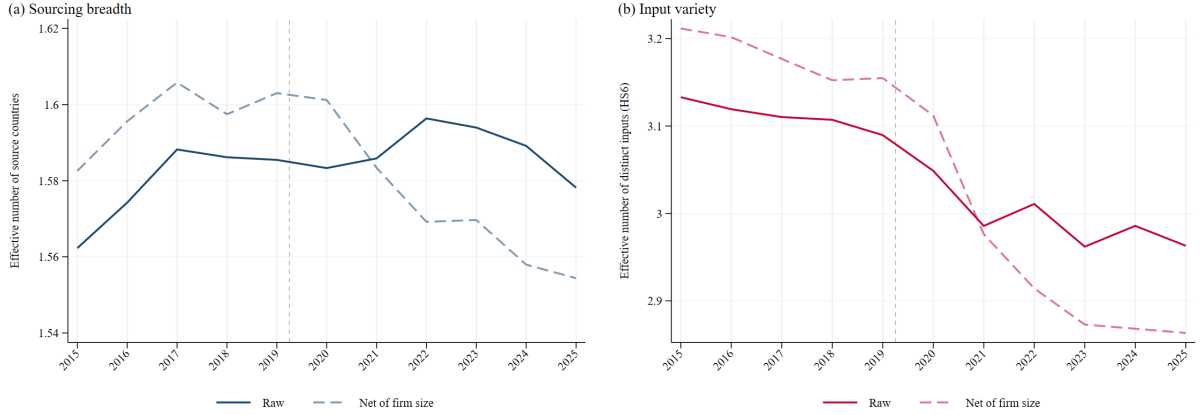


Figure 3: Number of firms importing intermediates.

Figure 5 disaggregates by sector, for the eight HS chapters accounting for the most importing firms. The cross-sector ordering of v_{ft} tracks technology rather than size: machinery (HS 84), vehicles (HS 87), precision instruments (HS 90) and electrical equipment (HS 85) show the highest input variety, and paper (HS 48) and plastics (HS 39) the lowest. This is the ordering one would expect if v_{ft} is a firm-level counterpart of the effective number of input types N_s , and **Appendix ?? reports the correlation with the input- output-based measure directly.** The declines in input variety are concentrated in precisely those high-variety sectors: HS 84 falls from about 3.97 to 3.48 and HS 87 from about 3.90 to 3.22 across the



Size-adjusted series are residuals from a regression of the measure on log total intermediate imports with sector effects, recentered at the sample mean. Year effects are deliberately excluded: with them the series is zero by construction. Dashed vertical line marks March 2020.
Source: Mexican customs microdata, Banco de Mexico Econlab.

Figure 4: Residualized effective number of source countries and distinct inputs.

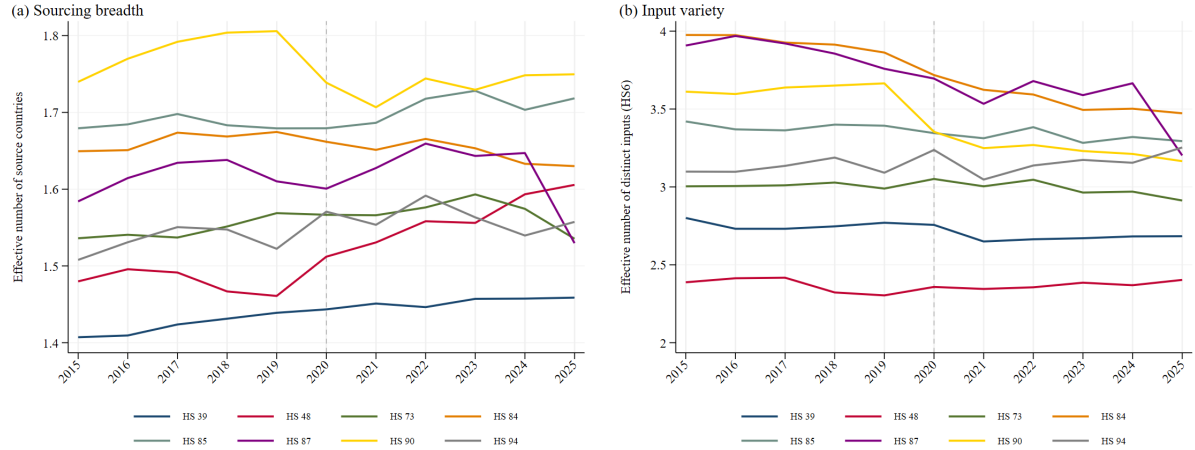
sample, while HS 48 is flat throughout. Sourcing breadth, by contrast, moves in no common direction: HS 48 rises after 2019 while HS 84 and HS 87 fall, so the flatness of the aggregate unweighted series conceals offsetting reallocation across sectors. Two cautions apply to this figure. Sector is assigned from the modal HS chapter of a firm’s pre-pandemic exports and held fixed, so a chapter label describes the firm’s output rather than the composition of the imports being measured. And the final year lies outside our estimation window: the sharp movements in HS 87 in 2025 coincide with the tariff changes discussed in Section 2.4.3, and we make no causal claim about them.

2.2.3 Internal dispersion and distance from home

Two geographic properties of a sourcing network matter in our model, and they are distinct. The first is how far apart the firm’s sources are *from one another*, which governs both the dissimilarity of their comparative advantage and the correlation of their shocks. The second is how far the sources are *from home*, which governs trade costs and is what the policy debate means by regionalization. A firm sourcing from Japan and Korea has low internal dispersion but might be distant from home; a firm sourcing from the United States and Chile has higher internal dispersion but might be closer.

We therefore construct

$$D_{ft} \equiv \sum_j \sum_{j'} s_{fjt} s_{fj't} d_{jj'} \quad \text{and} \quad \bar{d}_{ft}^H \equiv \sum_j s_{fjt} d_{0j} \quad (2.3)$$



Firm-year means within sector, for the sectors with the most importing firms. Sector is the modal HS chapter of the firm's pre-pandemic exports, held fixed. Vertical line marks 2020.
Source: Mexican customs microdata, Banco de Mexico Econlab.

Figure 5: Effective number of source countries and distinct inputs, by sector.

where $d_{jj'}$ is population-weighted bilateral distance and $j = 0$ denotes Mexico. D_{ft} is **Rao's quadratic entropy** with distance as the disparity metric, the canonical diversity-with-disparity index; $\bar{d}_{ft}^H$ is mean distance from home.

As robustness we report two alternative dispersion measures that are monotone in the sourcing set by construction: the length of the **minimum spanning tree** over the firm's sourcing locations, which has a natural interpretation as the cheapest network connecting the chain, and the **value-weighted radius of gyration around the firm's sourcing centroid**. Our results are not sensitive to the choice among the three.

We measure regional concentration by

$$H_{ft}^{\text{reg}} \equiv \sum_k \bar{\pi}_{fkt}^2, \quad \bar{\pi}_{fkt} = \sum_{j \in \mathcal{R}_k} s_{fjt} \quad (2.4)$$

together with the North American share $\bar{\pi}_{f,NA,t}$ and the share sourced from countries within a fixed radius of Mexico. We report all three because “regionalization” is used in the policy debate to mean at least these three different things, and they need not move together.

Figure 6 plots the two geographic measures of Equation (2.3). They behave differently, which is the first substantive return to separating them. Internal dispersion D_{ft} is close to flat: the unweighted mean sits near 1,800 km throughout, dips modestly in 2020–21, and ends the sample slightly below where it began. The value-weighted mean is roughly 1.8 times larger, near 3,300 km, and moves in the opposite direction after the pandemic,

rising to about 3,500 km in 2023–24. Distance from home does something else entirely. The unweighted mean of $\bar{d}_{ft}^H$ rises monotonically across the whole sample, from about 7,200 km in 2015 to about 8,400 km in 2025, with no visible break in 2020. Mexican importers have been sourcing steadily further from Mexico, during the pandemic as much as before it. Note also that the weighting reverses between the two panels: large importers have *more* internally dispersed chains than small ones but source *closer* to home, since the United States accounts for a far larger share of value than of firms. A measure of “how global” a supply chain is that does not distinguish these two objects will give a different answer depending on which one it happens to be tracking. As in Section 2.2.2, the within-firm series is nearly indistinguishable from the unweighted mean, so none of this is entry and exit.

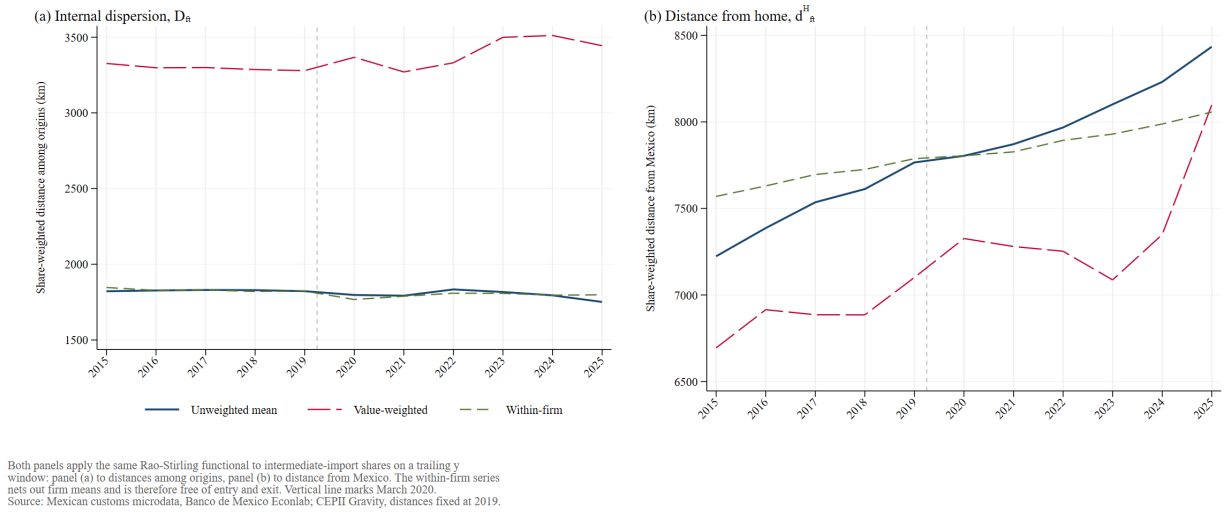


Figure 6: Geographic dispersion of supply chains.

Figure 7 makes the point directly. Across firms in 2019 the correlation between D_{ft} and $\bar{d}_{ft}^H$ is 0.00. Conditioning on how far a firm’s sources lie from one another says essentially nothing about how far they lie from Mexico. The structure visible in the scatter is not noise but geometry, and is worth stating because it clarifies what the measures do. A firm sourcing from two origins j and j' with shares s and $1 - s$ has $D_{ft} = 2s(1 - s)d_{jj'}$ and $\bar{d}_{ft}^H = s d_{0j} + (1 - s)d_{0j'}$; eliminating s traces a parabolic arc from $(0, d_{0j})$ to $(0, d_{0j'})$ with apex at $(d_{jj'}/2, (d_{0j} + d_{0j'})/2)$. Each visible arc is therefore a single pair of countries with the mixing weight varying across firms, and the dominant one—running from a mass of single-origin firms at $\bar{d}^H \approx 2,000$ km up to an apex near $(5,800, 7,800)$ —is the United States paired with China. The vertical mass at $D_{ft} = 0$ is the single-origin chains, which by construction have zero internal dispersion and which span the entire range of distance from home.

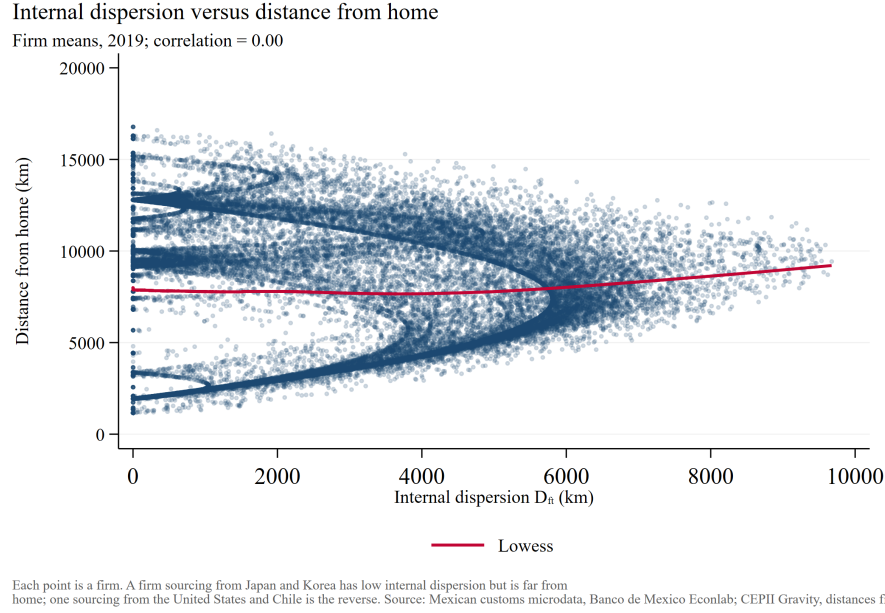


Figure 7: Correlation between the measures of geographic dispersion.

Figure 8 reports the three objects the policy debate calls regionalization. All three decline over the sample, and the decline is substantial: the value-weighted North American share of Mexican intermediate imports falls from about 0.54 in 2015 to about 0.41 in 2025, and the share sourced within 4,000 km of Mexico falls comparably. Within the estimation window the pandemic produces at most a pause: the North American share and the near-Mexico share both tick up between 2021 and 2023, by roughly one percentage point, before resuming their decline. More instructive is that the three measures do not move together. Between 2020 and 2022 the regional Herfindahl H_{ft}^{reg} rises while the North American share is flat, which can only mean that sourcing concentrated on a region other than North America. This is precisely why Section 2.2.3 reports all three: a paper that had documented only the Herfindahl would have described 2021–22 as regionalization, and one that had documented only the North American share would have described it as nothing at all. We emphasise that these are unconditional aggregates dominated by cross-firm composition and by the weight of the United States in Mexican import value, and that they are descriptive; the design in Section 2.4.3 is what tests whether firms reallocated in response to regional risk.

Figure 9 compares the three dispersion measures, each value-weighted and indexed to December 2019. Rao entropy and the radius of gyration are nearly indistinguishable, but their agreement is close to mechanical rather than corroborative: for a two-origin chain $D_{ft} = 2s(1-s)d_{jj'}$ and $\text{gyr}_{ft} = d_{jj'}\sqrt{s(1-s)}$, so one is a monotone transformation of the other, and with an effective chain size near 1.6 most firms are close to that case. The

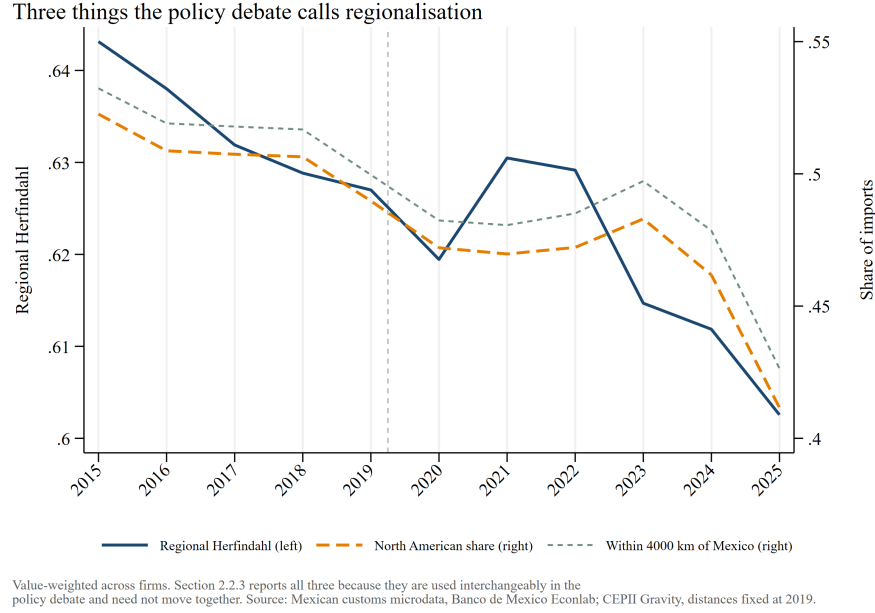


Figure 8: Measures of regional concentration of supply chains.

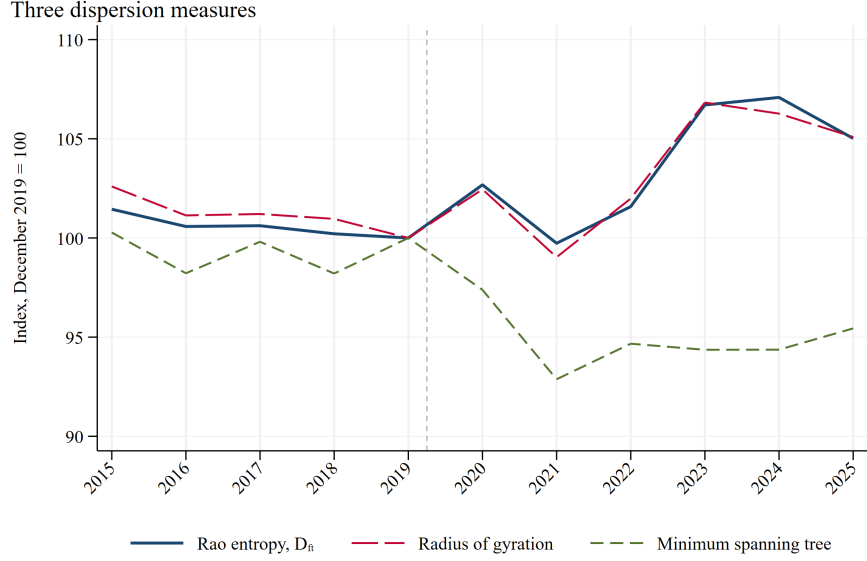
minimum spanning tree is the one measure of the three that is genuinely different, because it depends on the *set* of origins and not on the shares allocated across them, and it does not agree: it falls by about seven percent between 2019 and 2021 and remains four to five percent below its pre-pandemic level thereafter, while the two share-weighted measures end the period five percent above. The divergence is informative rather than troubling. It says that the extensive geographic footprint of the average chain contracted after 2020—origins were dropped, and the cheapest network connecting those that remained became shorter—while the value-weighted spread among the surviving origins widened.

2.2.4 Efficiency of a supply chain

Our measure of sourcing efficiency rests on the idea that a firm gains more from a supply chain whose members specialize in different products. Balassa (1965) defines the following index

$$\text{RCA}_{ip} \equiv \frac{X_{ip} / \sum_{p' \in \mathcal{P}} X_{ip'}}{\sum_{i' \in \mathcal{I}} X_{i'p} / \sum_{i' \in \mathcal{I}} \sum_{p' \in \mathcal{P}} X_{i'p'}}$$

which is bounded below by zero and unbounded above, making its distribution severely right-skewed. A pairwise dissimilarity built from squared differences in RCA *levels* is therefore dominated by a handful of products in which one country has an extreme value, which is a



Value-weighted and indexed to December 2019. The draft asserts results are not sensitive to the choice among the three; this figure is what establishes or refutes that. Source: Mexican customs microdata, Banco de Mexico Econlab; CEPII Gravity, distances f

Figure 9: Robustness of our measures of geographic dispersion of supply chains.

poor representation of the economic object we intend to measure.

We therefore construct our baseline measure of dissimilarity using the export-similarity index of Finger and Kreinin (1979), which is bounded and requires no transformation:

$$\text{Sim}_{ij} \equiv 1 - \sum_{p \in \mathcal{P}} \min \{ \omega_{ip}, \omega_{jp} \} \in [0, 1] \quad (2.5)$$

where $\omega_{ip} = X_{ip} / \sum_{p'} X_{ip'}$ is product p 's share of country i 's exports. As alternatives we report the same object built from the revealed symmetric comparative advantage transformation $\text{RSCA}_{ip} = (\text{RCA}_{ip} - 1) / (\text{RCA}_{ip} + 1) \in [-1, 1]$, and from the Euclidean distance in RCA levels. Products are defined at the HS 6-digit level throughout.

Supply chain efficiency is then defined as the value-weighted average dissimilarity among a firm's sources,

$$E_{ft} \equiv \sum_j \sum_{j'} s_{fjt} s_{fj't} \text{Sim}_{jj'} \quad (2.6)$$

which is again a **Rao–Stirling index**, with comparative advantage dissimilarity rather than distance as the disparity metric. Note that E_{ft} and D_{ft} are the same functional applied to two different disparity metrics, which makes the two forces in our trade-off directly comparable in magnitude.

Figure 10 plots E_{ft} , the value-weighted average comparative-advantage dissimilarity

among a firm's origins. The level is low: the unweighted mean sits near 0.115 against a theoretical maximum of one, and the value-weighted mean near 0.23. Both are far below one for the reason established in Section 2.2.2: with an effective chain size close to 1.6, a large share of firms source from what is effectively a single origin, and E_{ft} is zero for them by construction. The now-familiar size gradient reappears, and more sharply than for any other measure: weighting by import value doubles the index. The average dollar of Mexican intermediate imports flows through a chain whose members specialise in substantially more different things than the average firm's does. The two weightings also move differently. The unweighted mean is flat across the whole sample, varying within a band of about half a percentage point, while the value-weighted mean rises by roughly fifteen percent, from 0.222 in 2015 to 0.256 in 2025, with the increase concentrated after 2022. Large importers have been steadily increasing the comparative-advantage diversity of their sourcing, including through the pandemic. The within-firm series again lies almost on top of the unweighted mean, so this is not entry and exit.



E_{ft} is the Rao-Stirling index of equation (7): the share-weighted average dissimilarity of export baskets among a firm's origins, zero for a single-origin chain and rising as sources specialise in more different products. Dissimilarity is one minus the Finger-Kreinin similarity index and is held fixed, so movement here is movement in sourcing shares. Source: Mexican customs microdata, Banco de Mexico Econlab; Finger-Kreinin dissimilarity from BACI.

Figure 10: Average comparative-advantage dissimilarity of firms' sourcing strategies.

Before comparing E_{ft} with D_{ft} it is worth being explicit about what the two indices have in common, because it governs how the next two figures should be read. Both are Rao–Stirling quadratic forms in the same sourcing shares with a zero diagonal, and any such index factors exactly as

$$E_{ft} = \left(1 - \frac{1}{n_{ft}}\right) \bar{S}_{ft}, \quad D_{ft} = \left(1 - \frac{1}{n_{ft}}\right) \bar{d}_{ft} \quad (2.7)$$

where n_{ft} is the effective number of origins of Equation (2.1) and $\bar{S}_{ft}$, $\bar{d}_{ft}$ are the mean dissimilarity and mean distance among distinct origins under the normalised off-diagonal weights. The two measures therefore share a common factor that is a pure function of chain concentration, and a firm with one origin is at zero on both regardless of geography or specialisation. For the special case of a two-origin chain the point is starker still: $E_{ft}/D_{ft} = \text{Sim}_{jj'}/d_{jj'}$ exactly, independent of the shares.

That is what Figure 11 shows. The firm-level correlation between E_{ft} and D_{ft} is 0.86, and the visible fan of rays through the origin is precisely the family of two-origin chains, one ray per country pair with slope $\text{Sim}_{jj'}/d_{jj'}$. A correlation of this size should not be read as evidence that distance is a good proxy for comparative-advantage dissimilarity, because both measures are being driven up and down together by the concentration term in Equation (2.7). The informative object is the dispersion of the ray slopes, and Figure 12 isolates it by working at the country-pair level, where the concentration term is absent. There the weighted correlation between distance and dissimilarity is 0.12. The relationship has the sign the mechanism requires—the fitted line rises monotonically, from a dissimilarity of about 0.62 at zero distance to about 0.79 at twenty thousand kilometres—but it is shallow, and the residual variation is large: at any given distance, pairs range from roughly 0.2 to 1.0. Two features of the figure deserve comment. The dense cloud near the top, spanning all distances, is made up of pairs involving very small territories, which have almost nothing in common with anyone; they are numerous but carry no weight, which is why the figure is weighted by the value of Mexican imports flowing through each pair. More substantively, the heavily weighted pairs sit systematically *below* the fitted line, in a band between 0.35 and 0.6 and largely without regard to distance. The country pairs through which Mexican sourcing actually runs are more similar to one another than typical pairs at the same distance, so the comparative-advantage gain available to Mexican firms is smaller than the unweighted country-level relationship of Section 2.3 would suggest.

Figure 13 makes the same point in the time series, indexing both series to December 2019 because E is a share and D is measured in kilometres. If distance were a sufficient statistic for comparative-advantage dissimilarity the two lines would coincide. They do not. Between 2015 and 2019 efficiency rises by about five percent while internal dispersion falls slightly: firms were reallocating toward partners that were economically more different without becoming geographically more spread out. The two move together through the pandemic itself, rising in 2020 and falling in 2021, which is what one would expect if a common shock to sourcing shares drives both. They then separate again: dispersion rises sharply through 2023 and turns down, while efficiency continues to climb, and by the end of the sample the two have crossed. Taken with the country-pair evidence, the implication for the rest of the paper

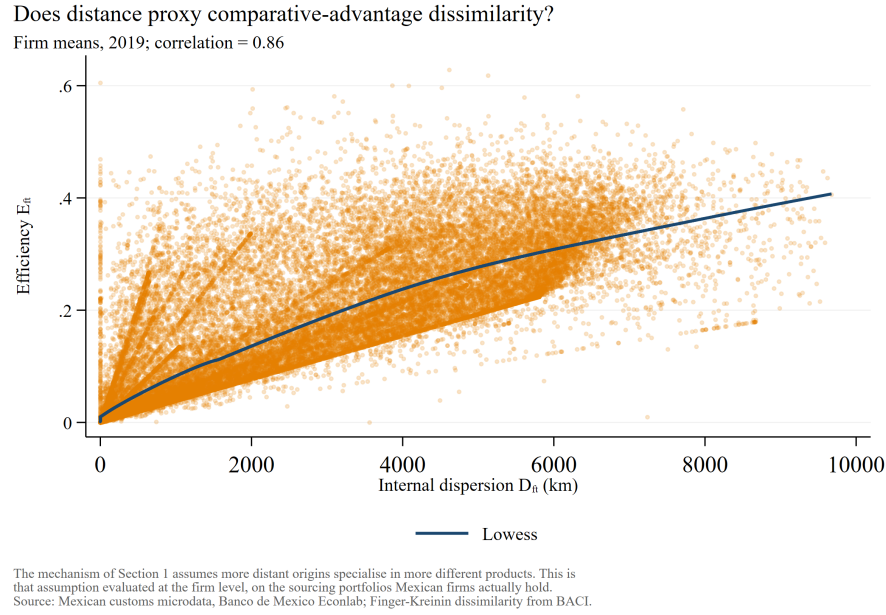


Figure 11: Firm-level efficiency vs. dispersion of sourcing strategies.

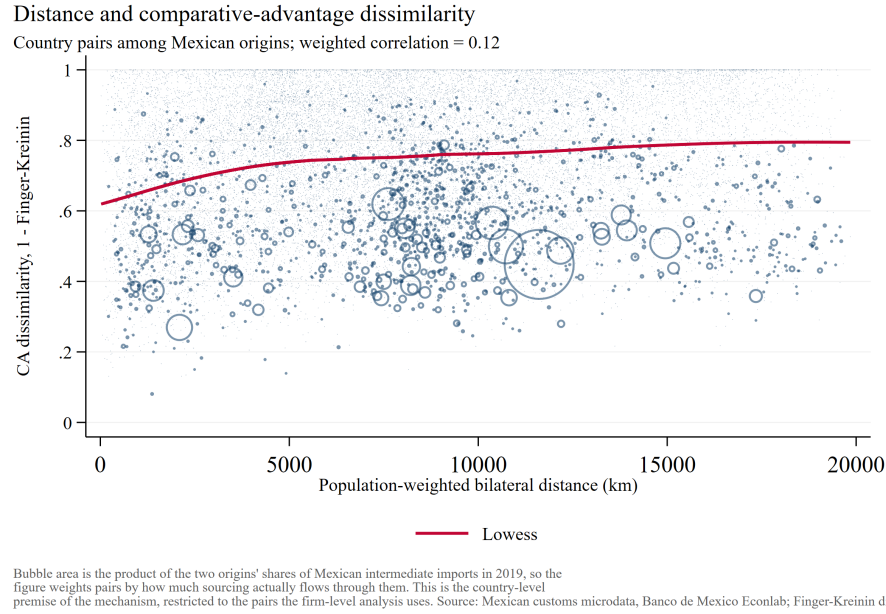


Figure 12: Firm-level distance vs. dissimilarity of sourcing strategies.

is that the efficiency and risk forces of Section 3.8 are not collinear in these data. Geography carries real but partial information about comparative advantage, and E_{ft} and D_{ft} must be carried separately rather than one standing in for the other. We note as elsewhere that the final year lies outside the estimation window and inside the altered tariff regime, and we

make no causal claim about it.

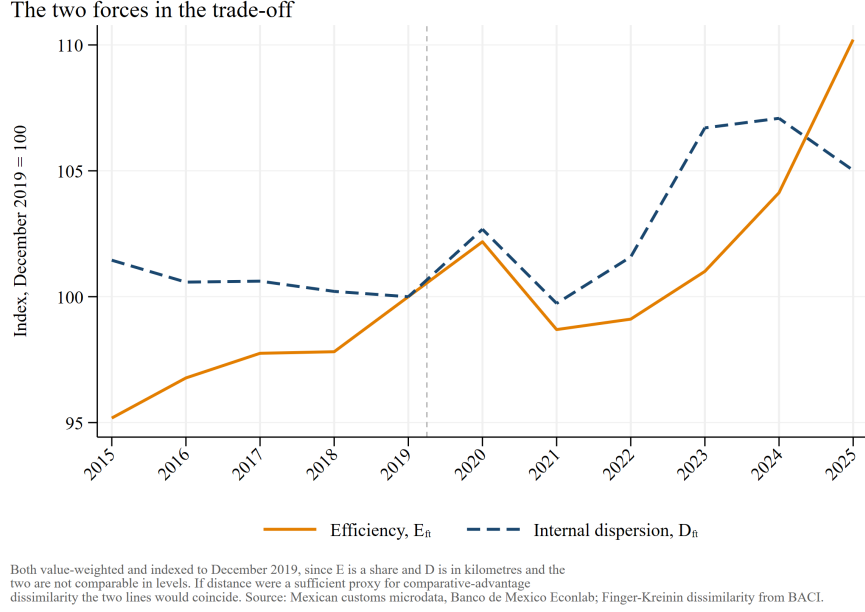


Figure 13: Average efficiency and dispersion of sourcing strategies.

2.2.5 Riskiness of a supply chain

Our model will imply a specific measure of the riskiness of a sourcing network. The covariance structure in Section 3.5 makes the object governing expected output losses a *portfolio variance* in the firm's sourcing shares. We construct its empirical counterpart directly.

Let $\rho_{jj'}$ denote the correlation of shocks in countries j and j' , with $\rho_{jj} = 1$, and let σ_j denote the standard deviation of country j 's shock. Define

$$V_{ft} \equiv \sum_j \sum_{j'} s_{fjt} s_{fj't} \sigma_j \sigma_{j'} \rho_{jj'}, \quad R_{ft} \equiv 1 - \frac{V_{ft}}{(\sum_j s_{fjt} \sigma_j)^2} \in [0, 1] \quad (2.8)$$

The ratio in (2.8) is the variance of the firm's shock portfolio relative to the variance it would face if all its sources were perfectly synchronized. It equals one when the firm sources from a single country or from perfectly correlated countries, and falls toward zero as sources become less synchronized. R_{ft} is therefore increasing in exposure to desynchronized shocks, which under input complementarity is the exposure that damages output. When all σ_j are equal, (2.8) reduces to one minus the share-weighted average pairwise correlation.

We construct $\rho_{jj'}$ and σ_j three ways: from the cyclical component of annual GDP following Hodrick and Prescott (1997); from monthly industrial production where available;

which addresses the poor small-sample properties of the Hodrick–Prescott filter noted by Hamilton (2018); and from the Oxford Stringency Index from March 2020 onward. The third is our preferred measure for the pandemic period because it is high-frequency and because it measures policy-driven supply interruption rather than output comovement. Correlations are averaged after a Fisher z transformation and back-transformed, since the arithmetic mean of correlation coefficients is biased toward zero.

Once the model of Section 3 is estimated, the factor structure in Assumption 2 delivers a structural measure of disruption probability, $\tilde{R}_{ft} = 1 - \mathbb{E}_{\mathcal{S}}[\mathcal{D}(\mathcal{S}; \mathcal{J}_{ft})]$, which incorporates input complementarity and redundancy in addition to correlation. We report the correlation between R_{ft} and $\tilde{R}_{ft}$ as a check on how much the reduced-form proxy misses.

2.2.6 Redundancy and exposure dispersion

The measures above treat a supply chain as a weighted list of countries. Our model implies that this is insufficient, because what determines fragility is how sourcing is organized across the distinct inputs the firm requires. We construct two additional sets of measures: the first ones capture redundancy *within* an input:

$$n_{ft}^{\text{src}} = \sum_p b_{fpt} \left(\sum_j (s_{fjt}^p)^2 \right)^{-1}, \quad \text{SS}_{ft} = \sum_p b_{fpt} \mathbb{1}\{|\{j : M_{fpjt} > 0\}| = 1\} \quad (2.9)$$

These are the value-weighted effective number of origins per product and the share of import value in single-sourced products. A computer requires one processor, but the assembler may buy processors from three suppliers in three countries; redundancy at the supplier level undoes fragility at the input level. Equation (2.9) is what distinguishes the two firms in the example of Section 2.2.1, and it is the margin along which our model predicts firms respond to *risk* rather than to cost.

The second one captures the evenness of regional exposure *across* inputs. Let $\pi_{fkt}^p = \sum_{j \in \mathcal{R}_k} s_{fjt}^p$. Define

$$\Delta_{ft} = \sum_p b_{fpt} \sum_k (\pi_{fkt}^p - \bar{\pi}_{fkt})^2 \quad (2.10)$$

Δ_{ft} is zero when every input is drawn from the same regional mix and is maximized when each input is drawn entirely from a distinct region. Crucially, it is orthogonal by construction to the firm’s aggregate geography: two firms with identical regional shares $\{\bar{\pi}_{fkt}\}$, identical effective chain size and identical average pairwise distance may differ arbitrarily in Δ_{ft} . Proposition 2 shows that Δ , and not geographic dispersion, is the object that input

complementarity penalizes: when losses are proportional across inputs, complementarity is irrelevant.

Figure 14 reports the two within-input measures of Equation (2.9), and the first thing they establish is that redundancy is overwhelmingly a large-firm phenomenon. The average Mexican importer holds 71 per cent of its intermediate import value in products it buys from a single origin, and its typical product is sourced from 1.20 effective origins. The average *dollar* of intermediate imports behaves quite differently: only about a fifth of it sits in single-sourced products, and its typical product draws on close to 1.9 effective origins. The gap between the two weightings is larger here than for any measure in Section 2.2, which is what one would expect if maintaining a second qualified supplier for a given input carries a fixed cost. It also means that the aggregate resilience of Mexican sourcing rests on a small number of firms: most importers have no within-input redundancy at all.

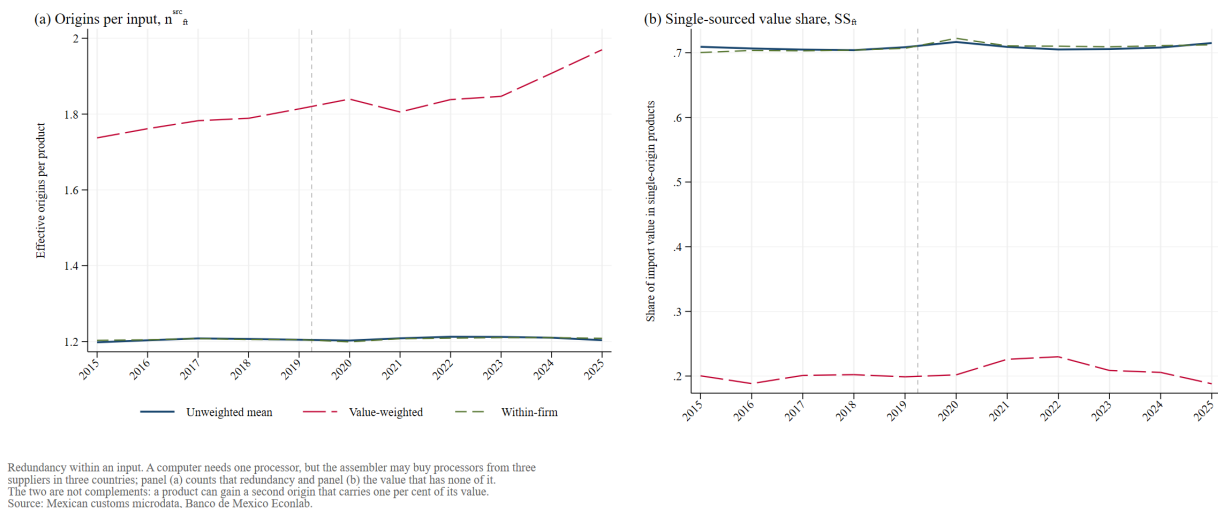


Figure 14: Input redundancy and single-sourced value share.

The two series also move differently over time, and not as mirror images. The value-weighted effective number of origins per product rises by roughly thirteen percent across the sample, from 1.74 to 1.97, interrupted only by a dip in 2021. The single-sourced value share does something else: it rises from about 0.20 in 2019 to 0.23 in 2022 before falling to 0.19 by the end of the sample. Large firms therefore concentrated *more* value in single-origin products through the disruption years while simultaneously adding origins to the products that already had more than one. The two measures are not complements and should not be read as such: n_{ft}^{sc} is share-weighted and responds to how evenly an input is split, while SS_{ft} is an extensive-margin count, so a product that acquires a second origin carrying one per

cent of its value leaves SS and moves n^{src} barely at all. As elsewhere, the unweighted and within-firm series are indistinguishable, so none of this is entry and exit.

Figure 15 plots exposure dispersion together with the two objects it is the difference between, $\Delta_{ft} = \sum_p b_{fpt} H_{fpt}^{\text{reg}} - H_{ft}^{\text{reg}}$. The levels are informative on their own. The average within-product regional Herfindahl sits near 0.78, an effective 1.3 regions per input, while the firm-level Herfindahl sits near 0.62, an effective 1.6 regions per firm. The typical Mexican importer therefore draws each individual input from what is close to a single region while spanning more than one region in aggregate: the configuration in which regional shocks hit some inputs and not others, which is precisely what input complementarity penalises according to our trade-off. Δ_{ft} itself is remarkably stable, moving in a band around 0.15 across eleven years, and the two Herfindahls drift down slowly and in parallel.

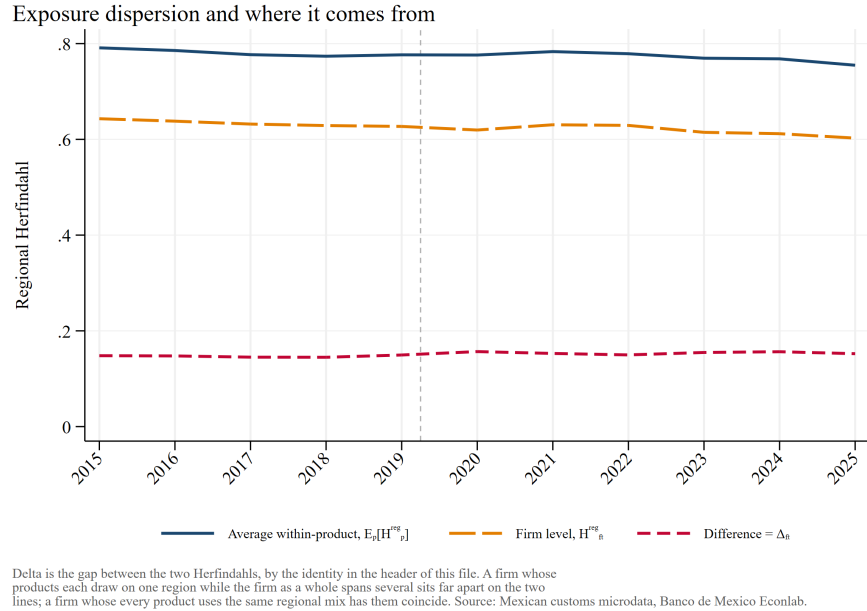


Figure 15: Evenness of regional exposure across inputs.

Exposure dispersion is not determined by a firm's aggregate geography. Two firms with identical regional shares $\{\bar{\pi}_{fkt}\}$ can differ arbitrarily in Δ_{ft} , which is what makes it a distinct object from H_{ft}^{reg} rather than a restatement of it. It is, however, bounded by that geography. Since $H_{fpt}^{\text{reg}} \leq 1$ for every product, the decomposition above implies

$$0 \leq \Delta_{ft} \leq 1 - H_{ft}^{\text{reg}} \quad (2.11)$$

so that a firm sourcing entirely within a single region necessarily has $\Delta_{ft} = 0$: the room a firm has to spread its exposure unevenly across inputs shrinks as its regional footprint

narrows.

Figure 16 plots the two against one another, and the scatter fills the triangle that (2.11) defines. Its upper envelope is the line $\Delta = 1 - H^{\text{reg}}$, attained by firms each of whose inputs is drawn from a single region; the mass along the horizontal axis is firms whose every input draws on the same regional mix; and the corner at $(1, 0)$ is firms confined to one region, for which the two coincide. The realised correlation of -0.83 therefore reflects this ceiling as much as any behavioural regularity, while the vertical spread within the triangle, which is wide at every value of H_{ft}^{reg} below unity, is the variation the measure is designed to isolate.

Where firms with different regional footprints are compared directly, we therefore also report the normalised index

$$\tilde{\Delta}_{ft} = \frac{\Delta_{ft}}{1 - H_{ft}^{\text{reg}}} \in [0, 1], \quad H_{ft}^{\text{reg}} < 1 \quad (2.12)$$

the share of the unevenness a firm could have exhibited, given its regional footprint, that it does exhibit. By construction $\tilde{\Delta}_{ft}$ is free of the dependence on aggregate geography that (2.11) imposes on Δ_{ft} , and it is defined for any firm sourcing from more than one region.

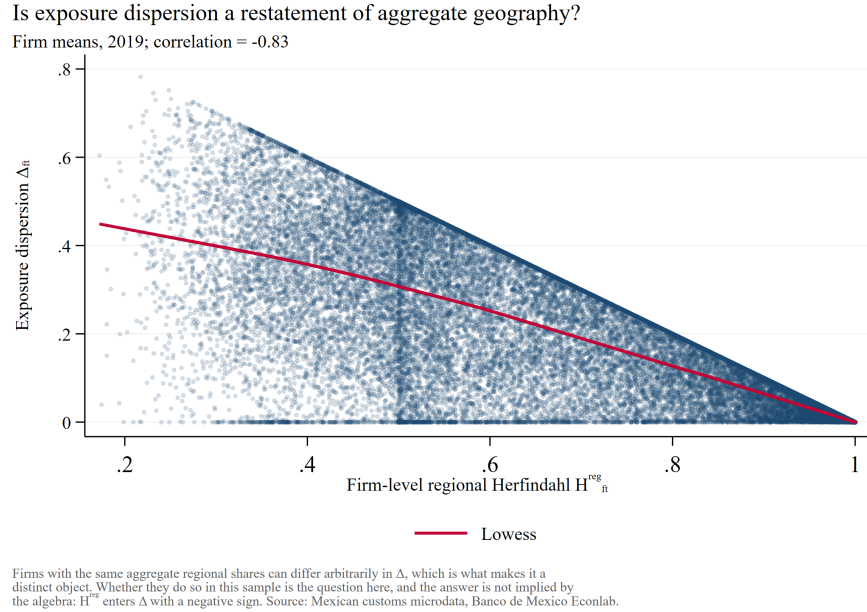


Figure 16: Exposure dispersion and the firm's regional footprint.

Figure 17 sets the two margins against one another, and it obliges us to restate Fact 4 of Section 2.3. The redundancy leg holds: within-input redundancy rises after 2020 and continues rising, ending some eight per cent above its 2019 level. The exposure-dispersion leg does not hold as previously stated. Δ does not fall after 2020; it *rises* sharply in 2020,

by close to five per cent, falls back to its 2019 level by 2022, and then rises again. Over the estimation window as a whole it ends above where it began. The pattern is nonetheless interpretable, and in a way that distinguishes the two channels Section 2.4.4 is at pains to separate. The 2020 spike is what mechanical disruption produces: when origins stop delivering, the inputs that lose an origin are left stranded on whichever region remains, and exposure becomes more uneven without any decision having been taken. The subsequent decline through 2022 is the reconfiguration, and it is the movement the model speaks to. We therefore state Fact 4 as: firms raised within-input redundancy steadily from 2020 onward, while exposure dispersion first rose mechanically with the disruption and then fell back over the two years that followed. The distinction between the two phases is not something the aggregate series can establish, and it is one of the things the design in Section 2.4.3 is meant to resolve.

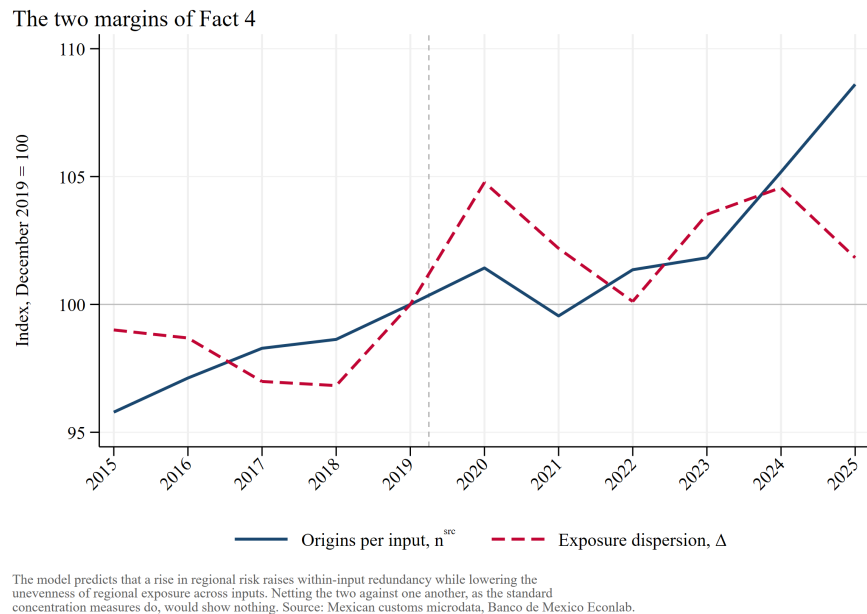


Figure 17: Origins per input and exposure dispersion.

2.2.7 Validating the proxies against realized outcomes

E_{ft} and R_{ft} are constructed using external data on comparative advantage and **business cycles**, and are therefore proxies for the efficiency and supply disruption that firms actually experience. An advantage of transaction-level monthly customs data is that both can be measured directly, which allows us to validate the proxies rather than assume them.

Customs records report both value and **quantity**, so we observe unit values at the firm $\times$

product $\times$ origin $\times$ month level. Let $\hat{u}_{fpt}$ be the residual from a regression of the log unit value paid by firm f for product p on product $\times$ month fixed effects, and define $U_{ft} = \sum_p b_{fpt} \hat{u}_{fpt}$: the average price the firm pays for its inputs relative to other importers of the same products in the same months. If E_{ft} captures comparative advantage gains, it should predict lower U_{ft} . We report this relationship. The usual caveat applies that unit values within an HS6 code conflate price with quality and composition, so we also report the relationship within narrow product-origin cells.

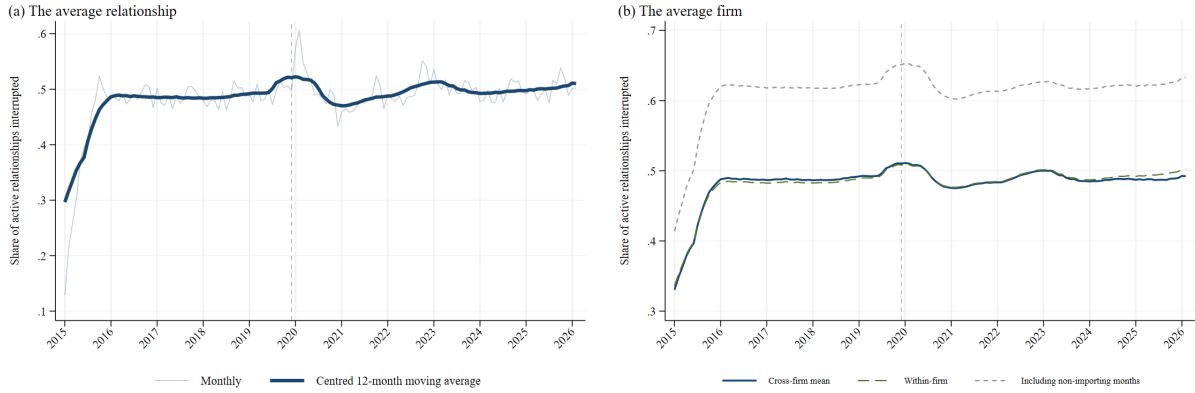
Monthly data allow us to observe supply interruptions directly. We classify a relationship (f, p, j) as *active* in month t if it recorded imports in at least three of months $t - 12$ through $t - 1$, and record an *interruption* when an active relationship records no imports in month t . The firm-level interruption rate is

$$I_{ft} = \frac{\#\{(p, j) \text{ active and interrupted in } t\}}{\#\{(p, j) \text{ active in } t\}} \quad (2.13)$$

This is a direct measure of the frequency of supply chain failure, constructed from the same data that generate our sourcing measures and requiring no external proxy. It is the empirical counterpart of $1 - \mathcal{D}$ in the model. We show that R_{ft} predicts I_{ft} , that the relationship is stronger in low- ζ_s sectors, and that I_{ft} spikes in 2020 in precisely the firm-months where our exposure measure predicts. To our knowledge this is the first direct measure of firm-level supply chain interruption frequency constructed from customs microdata, and it serves both as a validation of the risk index and as an outcome in its own right.

Figure 18 plots the realized interruption rate of Equation 2.13. Two features of its construction have to be read before its movements can be. The first is the opening year of the sample. A relationship enters the denominator only once it has accumulated a trading history, so at the start of the panel the activity rule can be satisfied only by the relationships that have been importing every month since the data begin, and those are precisely the ones least likely to record a gap. The rate rises from about 0.13 to about 0.49 over the first twelve months for this reason alone; the series is informative only once the lookback window lies entirely inside the sample. The second is the level at which it settles. Roughly half of all active supply relationships record no import in any given month, and that is a statement about ordering rhythm rather than about disruption. Firms place orders in lots, a relationship that ships every second month is entirely healthy, and at monthly frequency the measure counts the intervals between orders alongside genuine failures to deliver. What identifies disruption is therefore not the level of I_{ft} but its movement relative to a relationship's own normal cadence, which is how it enters the specifications below.

Read that way, the series behaves as the mechanism requires. The smoothed aggregate



A relationship is active in month t if it imported in at least 3 of months $t-12$ to $t-1$, and interrupted if it then records nothing in t . Panel (a) divides total interruptions by total active relationships; panel (b) averages firm-level rates, so it weights every firm equally. All series in panel (b) are centred twelve-month moving averages. The first two conditions on the firm importing something in t , so that firm exit is not counted as supply-chain interruption; the third does not, and the gap between them is firm dormancy.
Source: Mexican customs microdata, Banco de México Econlab.

Figure 18: Realized interruption rate over time.

rate is flat at 0.485 to 0.49 for the four years preceding the pandemic, rises to 0.52 across 2020 — the unsmoothed monthly series peaks just above 0.60 — and then *undershoots*, falling to 0.47 by early 2021 before returning to around 0.50 for the remainder of the sample. The undershoot is as informative as the spike: a period in which relationships delivered in more consecutive months than they ever had before is what inventory rebuilding looks like from the supplier side, and it is a reminder that I_{ft} measures the frequency of ordering as much as the reliability of it. Panel (b) establishes that none of this is composition. The cross-firm mean and the within-firm series — the same variable after firm means are removed — are visually indistinguishable throughout, so the aggregate movement is firms experiencing more interruptions rather than a changing mix of firms with different baseline rates. The third series in panel (b) retains the months in which a firm imports nothing at all, and sits about thirteen points higher, at 0.62. That gap is dormancy, not supply-chain failure: a firm that has stopped importing altogether registers an interruption on every one of its active relationships. Since the two series move together in 2020 and again in 2021, the disruption operated on both margins at once, and we keep them separate so that firm exit, which Section ?? treats as an outcome in its own right, is not counted twice.

Figure 19 disaggregates the same object by the origin region of the relationship, and the cross-sectional ordering is stable enough to be a fact about sourcing rather than a feature of 2020. Relationships with North American origins are interrupted least often, at about 0.475 before the pandemic, and those with Western European origins most often, at about 0.565, with East Asia, South and Southeast Asia, and Eastern Europe and Central Asia in

between. The ordering is emphatically *not* a distance ordering. North America is nearest to Mexico and is indeed the most continuous, but Western Europe is nearer than East Asia and is interrupted a full six points more often, and South and Southeast Asia, the most distant region in the group, is the second most continuous. What separates the regions is the kind of relationship they carry. Long-haul Asian sourcing in this sample is container-scale and repeated: large, regular, dedicated supplier links in assembly manufacturing that ship close to every month. European sourcing is tilted toward specialised machinery, chemical and instrument inputs bought in smaller and more occasional lots. Distance shapes what a firm buys from a region, and it is the resulting order frequency, not the shipping time, that sets the baseline rate. This is the same lesson as the geography panel of Table 1, where the firms sourcing furthest from Mexico were not the ones with the most concentrated chains: proximity, concentration and reliability are three distinct properties of a supply chain and the data decline to collapse them into one.

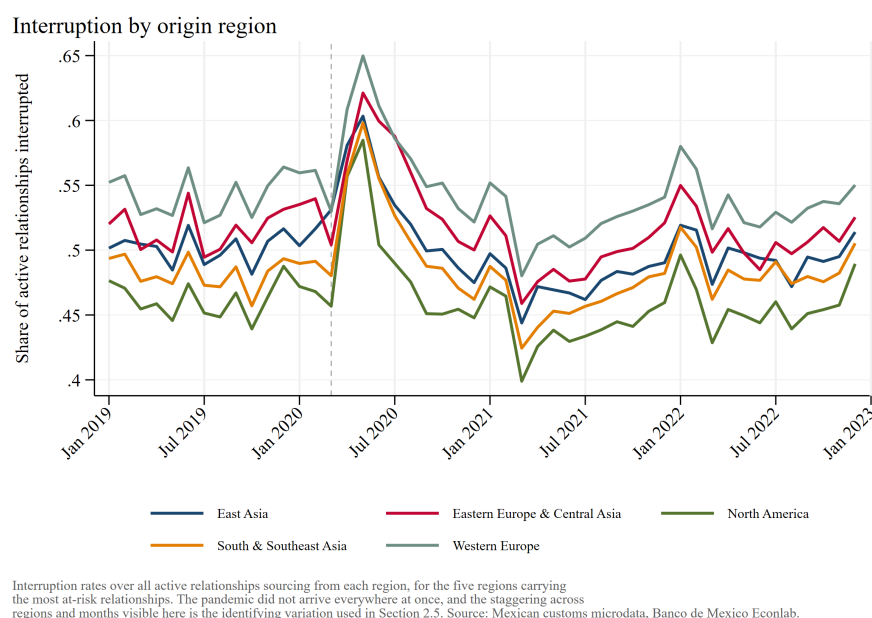


Figure 19: Realized interruption rate over time, by sourcing region.

The time-series behaviour by region qualifies how the pandemic variation can be used. The surge is common, not staggered: every region turns up sharply in the second quarter of 2020 and peaks in the same month, and every region undershoots its own pre-pandemic level in early 2021. What differs across regions is amplitude and persistence — North America and South and Southeast Asia rise most from the lowest bases and fall furthest in the 2021 trough, reaching 0.40 and 0.425 respectively against Western Europe’s 0.48 — and the level ordering survives the episode intact. Identification in Section 2.4.3 therefore cannot

rest on regions being hit at different *dates*; it rests on firms with otherwise similar chains being differently *exposed* to regions whose disruptions differed in size, which is what the exposure measures of Section 2.2 are built to capture. That I_{ft} moves at all, in the right direction, at the right time, and by an amount that varies systematically with where a firm sources, is what licenses using those measures as proxies for a delivery probability that is never observed directly. The figure also makes plain why the raw monthly series is unusable without adjustment: the January peaks recur in every year of the sample, and only the 2020 surge and the 2021 trough clear that seasonal band.

2.2.8 Persistence

~~Finally, we report the evolution of our measures through May 2026, outside the estimation window. This period includes a materially changed tariff environment and we make no causal claims about it. Its purpose is to establish whether the reconfiguration documented through 2023 persisted, reversed, or accelerated. We report the raw series alongside the fitted path implied by extrapolating our estimates from the main sample, and we discuss the divergence between them in light of the policy changes described in [Appendix ??](#).~~

2.2.9 The measures in the estimation sample

Table 1 describes the estimation sample. The unit of observation is a firm-year, and the sample is the 115,533 Mexican firms that import intermediate goods at some point between 2015 and 2023, contributing 462,907 firm-years in total. The four panels follow the construction in Section 2.2: how large a sourcing network is, how it is arranged in space, how much comparative-advantage gain it captures, and how it is organised across the distinct inputs the firm buys. The second pair of columns restricts to firm-years with more than one effective origin. This is not a robustness sample but a necessary one: several of the measures are degenerate for a firm that sources everything from a single country, which by construction has $D_{ft} = E_{ft} = \Delta_{ft} = 0$ regardless of where that country is, and 37 percent of firm-years are of this kind. The restricted columns therefore show the distribution among the firms for which the structural measures carry information, and the difference between the two pairs is itself informative about how much of the sample the single-origin case accounts for.

The first thing the table establishes is how far a firm’s sourcing network is from the picture of a widely distributed global value chain. The average importer buys from 5.18 countries but has an effective number of origins of 1.58, and buys 21.28 distinct HS6 products with an effective input variety of 3.06. The gap between the raw and effective counts is a factor of three for origins and of seven for products: the long tail of small shipments that dominates

the raw counts carries almost no expenditure. The standard deviations exceed the means for both raw counts, so these distributions are heavily right-skewed and the means understate how concentrated the typical firm is, which is one reason the specifications below work with logs and with size-residualised measures rather than levels.

More important for the mechanism is what happens *within* inputs. The average firm holds 70.8 percent of its intermediate import value in products it buys from exactly one country, and its typical product is sourced from 1.21 effective origins. Conditioning on firms that use more than one origin barely changes this: they still single-source 53.0 percent of their value, with 1.33 effective origins per product. Multi-origin sourcing is therefore not the same thing as redundant sourcing. A firm with 1.94 effective origins that assigns each of its inputs to one of them is diversified at the level of the firm and undiversified at the level of every input it depends on, which is precisely the configuration Proposition 2 identifies as fragile: complementary inputs, each hanging on a single origin, so that a regional disturbance removes some inputs entirely rather than reducing all of them proportionally. Exposure dispersion averages 0.118 across all firms and 0.189 among multi-origin firms, against a maximum of $1 - H_{ft}^{\text{reg}}$ by Equation (2.11), so firms sit roughly a third of the way toward the most uneven configuration their regional footprint permits.

The geographic panel qualifies the reading of concentration as proximity. The average regional Herfindahl of 0.809 implies an effective 1.24 regions per firm, and 1.44 among multi-origin firms, so even firms spanning several countries rarely span several regions. North America accounts for 39.2 per cent of the average firm’s intermediate imports, with a standard deviation of 0.421 on a variable bounded in $[0, 1]$, indicating a distribution concentrated near its two endpoints rather than around its mean. Distance from home, however, is essentially flat across the two samples and slightly *higher* for firms with a single origin: 7,694 km on average against 7,568 km for multi-origin firms, which implies a mean near 7,900 km for the single-origin group. Firms with no diversification at all are thus the ones sourcing furthest from Mexico, not the ones sourcing nearest to it. Sourcing concentration and sourcing proximity are separate properties of a chain, and any account of regionalisation that treats them as one will misread which firms are exposed.

Finally, the outcome rows show one half of the trade-off unconditionally and not the other. Firms with more than one effective origin export more, with mean log exports of 12.75 against 12.51, and they also capture more comparative-advantage gain, with E_{ft} of 0.186 against 0.116. That is the efficiency leg of the mechanism, and it appears in the raw cross-section. The coefficient of variation of exports, by contrast, is marginally *lower* for multi-origin firms, 1.408 against 1.452, which is the opposite of what the risk leg predicts. We do not read this as evidence against the mechanism, because the comparison is confounded

in an obvious direction: single-origin firms are small, and small firms have lumpy monthly exports for reasons that have nothing to do with the geography of their suppliers. The average firm’s monthly exports have a coefficient of variation above one, which gives some sense of how much of this variation is scale and lumpiness rather than supply-chain structure. Establishing the risk leg therefore requires holding size and sector fixed, which Section 2.3.2 does, and ultimately requires variation in exposure that is not chosen by the firm, which is what Section 2.4.3 supplies.

2.3 Evidence on the trade-off

Our mechanism requires that geographic distance be associated with more dissimilar comparative advantage and with less synchronized supply shocks. We document both at the aggregate and firm-level.

2.3.1 Aggregate-level evidence

Figure 20 plots the relationship between bilateral distance and comparative advantage dissimilarity across all 12,403 country pairs in our sample. Countries further apart specialize in more distinct sets of products. Figure 21 plots the relationship between distance and the correlation of cyclical output components: countries further apart have less synchronized business cycles. Figure 22 combines the two, showing directly the negative relationship between comparative advantage dissimilarity and shock synchronization that generates the trade-off. Because the cyclical-correlation measure is estimated on short annual series, we reproduce all three figures using the pairwise correlation of the Oxford Stringency Index from March 2020 onward (Figures ?? and ??). The pattern is sharper in the stringency data, which is what one would expect if the lockdown-driven component of the pandemic is a cleaner measure of correlated supply disruption than filtered annual GDP.

These relationships are what give geography its role in the model of Section 3: they justify parameterizing both the within-region correlation of cost draws, χ_k , and the regional structure of delivery shocks in terms of the same geographic partition.

We next document the aggregate evolution of the measures in Section 2.2. This is shown in Figures 23 and 24.

Four facts stand out, of which the last is new and, in our view, the most important. The first two concern the aggregate extent of sourcing. Between 2014 and 2019, Mexican firms sourced from a growing effective number of countries, and from countries with increasingly dissimilar comparative advantage; the two series rise roughly in parallel. Both then fall sharply in 2020, returning to approximately their 2014 levels, and the recovery is

	All firms		$n_{ft} > 1$	
	Mean	S.D.	Mean	S.D.
<i>Extent</i>				
Origin countries (raw count)	5.18	7.65	7.72	8.77
Effective no. of origins, n_{ft}	1.58	0.91	1.94	1.00
Distinct HS6 products (raw count)	21.28	46.16	32.11	55.61
Effective input variety, v_{ft}	3.06	3.29	3.96	3.79
<i>Geography</i>				
Internal dispersion (km), D_{ft}	1,819	2,236	2,917	2,194
Distance from home (km), $\bar{d}_{ft}^H$	7,694	4,290	7,568	3,799
Regional Herfindahl, H_{ft}^{reg}	0.809	0.229	0.694	0.221
North American share	0.392	0.421	0.402	0.378
<i>Efficiency and risk</i>				
Efficiency, E_{ft}	0.116	0.133	0.186	0.124
Risk, R_{ft}				
Realized interruption rate, I_{ft}	0.620	0.227	0.621	0.214
Relative unit value paid, U_{ft}				
<i>Structure</i>				
Effective origins per product, n_{ft}^{src}	1.21	0.43	1.33	0.50
Single-sourced value share, SS_{ft}	0.708	0.377	0.530	0.381
Exposure dispersion, Δ_{ft}	0.118	0.166	0.189	0.175
<i>Technology and outcomes</i>				
Input substitution elasticity, ζ_s	0.347	0.218	0.358	0.218
Effective input types, N_s				
Log exports	12.50	3.40	12.75	3.47
Coefficient of variation of exports	1.454	1.101	1.408	1.108
Observations	473,536		288,165	
Number of firms	115,533		66,006	

Notes. Firm-year averages, 2015–2023, over intermediate-good imports. Definitions in Section 2.2. ζ_s is taken from Gosselin (2025), whose industry-level estimates are mapped from BEA summary industries to HS chapters through the BEA–NAICS and NAICS–HS concordances; a firm is assigned the value of its own sector. N_s is sector-level and reported as a firm-weighted average. Because ζ_s and N_s are constant within a sector, the dispersion reported for them is between sectors rather than between firms. I_{ft} is the share of a firm’s active supplier relationships—those recording imports in at least 3 of the preceding 12 months—that record no import in a given month, evaluated only in months in which the firm imports something, so that exit is not counted as interruption, and aggregated to the firm-year by summing interruptions and relationships at risk rather than by averaging period rates. Because firms order in lots, close to half of all active relationships are idle in any month even in the years before 2020: the level reflects ordering frequency, and it is the movement of I_{ft} rather than its level that measures disruption. Log exports and the coefficient of variation are defined for firms with positive exports in the year; the coefficient of variation is taken across the months of the year lying inside the firm’s export span, so that entry and exit do not enter it.

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Table 1: Summary statistics, main estimation sample

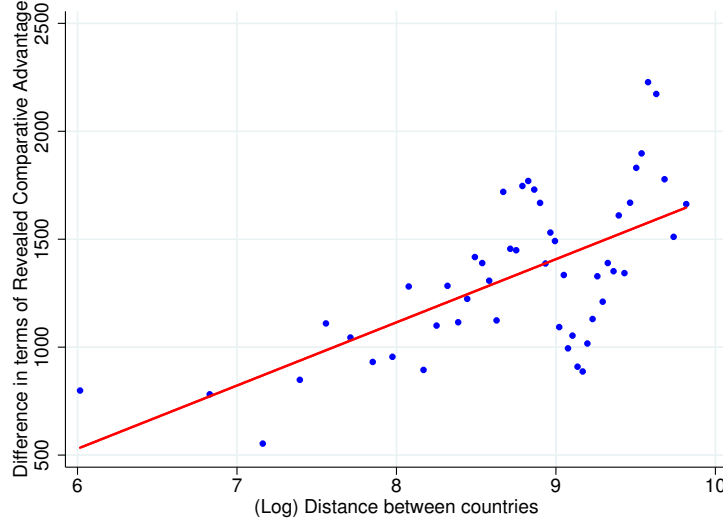


Figure 20: Geographical distance and differences in RCA

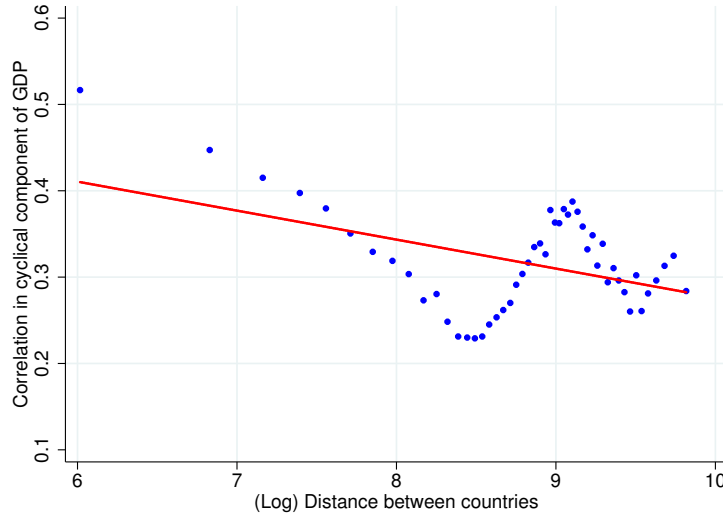


Figure 21: Geographical distance and cyclical GDP correlation

incomplete, since neither regains its 2019 level within the sample. The third fact concerns geography rather than extent. The North American share of Mexican intermediate imports rises after 2020, and the regional Herfindahl H_{ft}^{reg} rises with it. Taken alone, this is consistent with the policy narrative of nearshoring, but it is equally consistent with several alternative explanations that we address in Section 2.4.4.

The fourth fact is that this apparent concentration conceals two movements in opposite directions. Exposure dispersion Δ_{ft} *falls* after 2020 while the effective number of origins per product, n_{ft}^{src} , *rises*. Firms became more concentrated in the sense that their different inputs

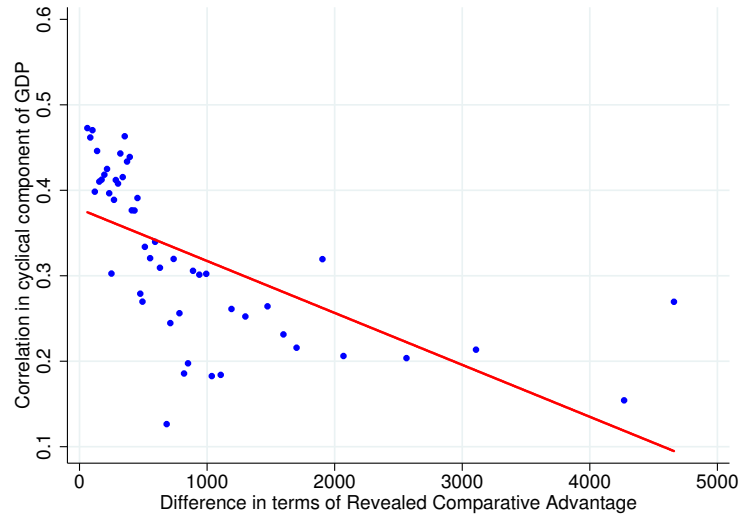


Figure 22: Comparative Advantage and cyclical GDP correlation

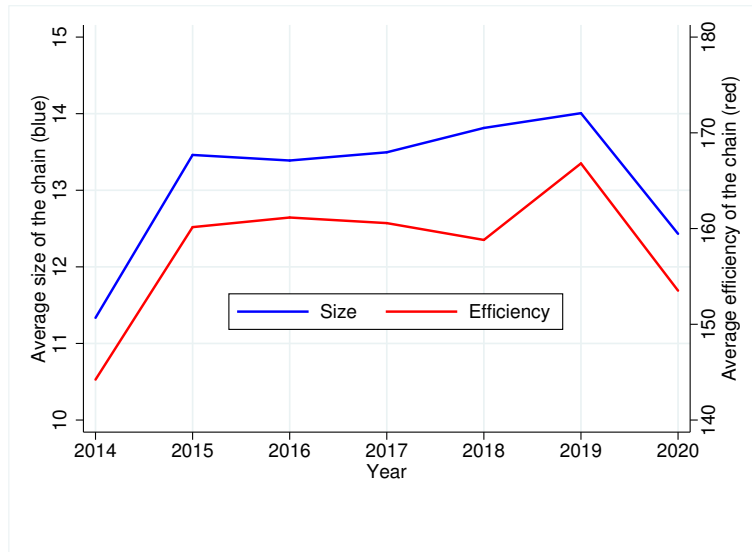


Figure 23: Average size and efficiency of supply chains

are increasingly drawn from a common regional mix, and simultaneously less concentrated in the sense that each individual input is increasingly sourced from more than one country. This is precisely the pattern our model predicts, and it is invisible to the concentration measures used in the existing literature, which net the two margins against one another. It is also why we do not frame the empirical question as “did supply chains concentrate?” The substantive question is concentration along which margin, and **the two margins carry opposite signs**.

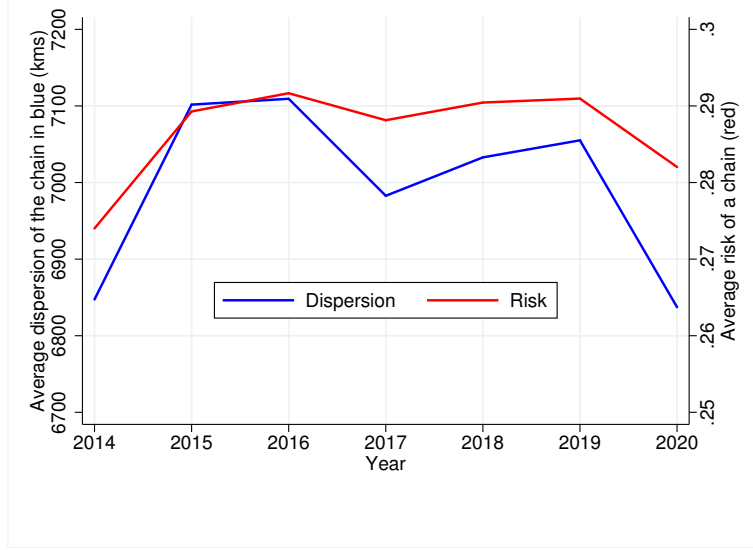


Figure 24: Average dispersion and risk of supply chains

2.3.2 Firm-level evidence

Before turning to identification, we establish the cross-sectional relationships that motivate the exercise. We estimate

$$y_{ft} = \beta_0 + \beta_1 R_{ft} + \beta_2 (1 - \zeta_{s(f)}) + \beta_3 R_{ft} \times (1 - \zeta_{s(f)}) + \mathbf{x}'_{ft} \boldsymbol{\gamma} + \epsilon_{ft} \quad (2.14)$$

where y_{ft} is the log of exports or the coefficient of variation of exports, and $\mathbf{x}_{ft}$ contains firm size and sector and year fixed effects. Standard errors are clustered at the firm level. The specification includes an interaction with $1 - \zeta_s$, a direct estimate of input complementarity. Our mechanism predicts $\beta_3 > 0$, and predicts that the trade-off vanishes as $\zeta_s \rightarrow 1$. A null result in high- ζ_s sectors is therefore informative rather than uninformative. We report the specification in terciles of ζ_s as well as continuously.

We emphasize that (2.14) is descriptive. R_{ft} is an equilibrium object: firms that source from desynchronized locations do so for reasons that also predict their performance. The remainder of this section is devoted to identification.

2.4 The effects of increased supply risk

We use the COVID-19 pandemic as our source of variation. Three features make it uniquely suited to our mechanism.

First, it was a shock to the *correlation structure* of supply disruptions, not merely to their level. Our model's comparative statics concern $\{r_k\}$, the probabilities of region-specific

disruption, and the pandemic moved these differentially across regions. Episodes such as the 2011 Tōhoku earthquake or the 2018–19 tariff escalation shifted risk in a single location or a single country and cannot identify a response to regional structure.

Second, its incidence was *staggered across regions*. East Asian supply was disrupted in the first quarter of 2020 while European and American supply was not; European supply was disrupted in the second quarter as East Asian supply recovered; parts of Southeast Asia experienced their most severe disruption in mid-2021, long after North America had reopened. A firm importing one input from Guangdong and another from Bavaria was disrupted at different times for different inputs. This staggering is the identifying variation we exploit, and it is lost if the pandemic is collapsed into a single post-2019 indicator.

Third, it was severe enough to plausibly shift beliefs, which is the object our model’s comparative static concerns. We return to this interpretive point in Section 2.4.4.

2.4.1 Measuring firm-level exposure to supply disruptions

We construct exposure at the firm $\times$ product level from pre-determined sourcing shares interacted with realized disruption in the source country. Let $s_{fj,2019}^p$ denote firm f ’s average share of input- p imports sourced from country j over 2017–2019, fixed thereafter. Our baseline monthly exposure is

$$\text{Exp}_{fpt} = \sum_j s_{fj,2019}^p \text{Str}_{jt} \quad (2.15)$$

where Str_{jt} is the **Oxford Stringency Index** in country j in month t , normalized to zero in the pre-period. We also report results using a cumulative exposure over the first twelve months of the pandemic, $\text{Exp}_{fp} = \sum_{t \in T_0} \text{Exp}_{fpt}$, which is the object used in the local projections below.

The stringency index measures government policy rather than realized supply failure. We therefore construct an alternative shock measure directly from trade flows: for each origin country j , product p and month t , we compute the log change in j ’s exports of p to the *world excluding Mexico*, relative to the same month in 2019,

$$\text{Shock}_{jpt} = -\Delta \log X_{jpt}^{\text{LO}} \quad (2.16)$$

and substitute it into (2.15). The leave-out construction ensures that the measure is not mechanically driven by Mexican demand. In our preferred specification we instrument (2.16) with **stringency**, which isolates the policy-driven component of realized supply failure.

2.4.2 First stage specification

Before turning to reconfiguration, we verify that the exposure measure captures actual disruption. We estimate, on the firm $\times$ product $\times$ origin $\times$ month panel,

$$\log M_{fpjt} = \beta \text{Str}_{jt} + \alpha_{fpj} + \delta_{ft} + \gamma_{pt} + \varepsilon_{fpjt} \quad (2.17)$$

The relationship-level fixed effect α_{fpj} absorbs all time-invariant determinants of a given sourcing relationship; δ_{ft} absorbs everything happening to the firm in that month, including its own lockdown exposure and its demand conditions; γ_{pt} absorbs global product-specific shocks such as the semiconductor shortage. The coefficient β is identified from variation in **lockdown severity** across the origins from which the *same firm* sources the *same product* in the *same month*. We expect $\beta < 0$.

2.4.3 Second stage specification

Our main specification is estimated on the firm $\times$ product $\times$ month panel:

$$y_{fpt} = \sum_{\tau \neq \tau_0} \beta_{\tau} \text{Exp}_{fp} \times \mathbb{1}\{t = \tau\} + \alpha_{fp} + \delta_{ft} + \gamma_{pt} + \varepsilon_{fpt} \quad (2.18)$$

with τ_0 = December 2019 as the omitted period. The outcomes y_{fpt} are the product-level analogues of the measures in Section 2.2: the number and effective number of origins for the product, an indicator for single sourcing, the regional Herfindahl of the product’s origins, the North American share, and indicators for the addition of a new origin and for the permanent termination of an existing origin, where “permanent” is defined as absence for twenty-four consecutive months.

The identifying assumption is that, conditional on firm $\times$ month and product $\times$ month effects, the pre-pandemic geographic composition of a firm’s sourcing of product p is unrelated to shocks that would have differentially affected the sourcing of p within that firm after 2019. The firm $\times$ month effects are what make this credible. They absorb, without requiring us to model them: the collapse in US demand for Mexican exports, the Mexican domestic lockdown, firm-level credit conditions, managerial quality, firm size and growth, sector, and IMMEX status. What remains is variation generated by the fact that different inputs of the same firm were sourced from places that locked down at different times.

Because pandemic incidence was staggered and outcomes adjust slowly, we also report Jordà local projections (Jordà 2005),

$$y_{fp,t_0+h} - y_{fp,t_0-1} = \beta_h \text{Exp}_{fp} + \delta_{f,t_0+h} + \gamma_{p,t_0+h} + \varepsilon_{fp,h}, \quad h = 0, \dots, 48, \quad (2.19)$$

which trace the horizon at which reconfiguration occurs. Local projections are preferable here to a fixed post-period because the relevant adjustment margin, the formation and severance of supplier relationships, operates over years rather than months, and because they impose no restriction on the shape of the dynamic response. Standard errors are two-way clustered by firm and by origin-country-of-2019-share, following the shift-share logic discussed below.

The design in (2.18) identifies the reallocation of sourcing *within* firms. To speak to aggregate regionalization we also estimate a firm-level version with $\text{Exp}_f = \sum_p b_{fp,2019} \text{Exp}_{fp}$. This is a shift-share design in which the shares—pre-period sourcing weights—are plainly endogenous. We therefore rely on the exogeneity of the shocks rather than of the shares, following Borusyak et al. (2022): we exploit the large number of origin-month shocks, and **we recenter by including the expected exposure implied by the distribution of shocks** as a control. Because the treatment is continuous, and continuous-dose difference-in-differences identifies a causal parameter only under assumptions that rule out selection on gains (Callaway et al. forthcoming), we report as our headline firm-level result a binarized specification comparing the top and bottom terciles of Exp_f , with the continuous specification as a robustness check.

Our estimation sample ends in December 2023. This reflects a deliberate trade-off. A longer window would capture more adjustment, but from March 2025 onward Mexican firms operated under a substantially altered tariff regime, and any regionalization observed thereafter cannot be cleanly attributed to supply risk. Ending in 2023 provides around three years of post-shock adjustment, which our local projections indicate is sufficient to capture the bulk of the response, while stopping before the tariff regime changed. Section 2.2.8 reports the full series through May 2026 as a descriptive persistence check, with the confounds explicitly flagged.

2.4.4 Threats to identification

Several features of our setting could produce the patterns we document through channels other than the one we propose, and we take them in turn.

The most serious is the transition to USMCA, which entered into force in July 2020, in the middle of our treatment window, and whose automotive regional value content requirement of 75% phased in over our post-period. A headline finding of North American regionalization is exactly what a binding rules-of-origin requirement would produce, so we address it in three ways. First, we use a **product-level measure of RVC exposure**, indicating

whether HS6 product p is an input into exports that claim USMCA preference, and include it interacted with time. Second, we report results excluding the automotive complex entirely. Third, we use the automotive sector as a *positive* placebo. It is where the policy channel should be strongest, so a similar effect size inside and outside autos would indicate that our estimates are contaminated, while a larger effect in autos that survives the RVC control is informative about the interaction of policy and risk. A related competing channel operates through freight rather than policy: the 2021–22 container crisis raised shipping costs disproportionately on long-haul routes, which pushes toward regionalization for reasons of cost rather than risk. Where **route-level freight data** are available we control for them directly, but the more fundamental response is that our model distinguishes the two channels. A cost shock operates through $C(\mathcal{J})$ and predicts substitution toward the cheapest alternative, whereas a risk shock operates through $\Lambda(\mathcal{J})$ and predicts substitution toward alternatives *synchronized with the firm’s remaining sources*, together with an increase in within-input redundancy. We test this distinction directly in Section 2.4.5.

A second set of concerns is internal to the outcome variable rather than external to it. In 2020, imports from locked-down origins fell simply because shipments did not occur, which is disruption rather than reconfiguration. We separate the two by excluding 2020 from the outcome window in our main specifications, using it only to estimate (2.17), and by **measuring outcomes at the relationship extensive margin with a twenty-four-month absence criterion**. Local projections make the distinction visible, since a purely mechanical response reverts to zero while reconfiguration does not. Mean reversion raises a similar issue, in that firms whose 2019 sourcing of a product was unusually concentrated will drift toward the mean regardless of treatment. We therefore control for the 2019 level of the outcome interacted with time, and report long-difference specifications as a robustness check.

The composition of the estimation sample poses two further problems. Firms that exited in 2020 were plausibly the most exposed, so we report results on a balanced panel and on the full panel, estimate exit as an outcome in its own right, and report continuing relationships separately from the full set including entry. The second problem is specific to the Mexican setting, since sourcing decisions at temporary-import establishments may be made abroad rather than by the Mexican entity. We therefore report all results separately by IMMEX status, and note that under our interpretation the response should be attenuated among IMMEX firms, whose Mexican entities might not be the ones choosing their suppliers.

Finally, a point of interpretation. Our model’s comparative static concerns a change in perceived regional risk, r_k , while our treatment is realized disruption, and the identifying logic is that firms disrupted more severely updated their beliefs more. This is an assumption rather than a result, and we state it as such. The heterogeneity test in Section 2.4.5 on

responses to *later* disruptions provides indirect evidence for it.

2.4.5 Falsification and heterogeneity

Our model generates sign restrictions that a purely reduced-form analysis cannot, and we use them as tests rather than as descriptive heterogeneity.

The sharpest of these concerns input complementarity. Proposition 1 implies that the response should vanish as $\zeta_s \rightarrow 1$, which we test by estimating

$$y_{fpt} = \sum_{\tau} [\beta_{\tau} + \psi_{\tau} (1 - \zeta_{s(f)})] \text{Exp}_{fp} \times \mathbb{1}\{t = \tau\} + \alpha_{fp} + \delta_{ft} + \gamma_{pt} + \varepsilon_{fpt} \quad (2.20)$$

and predicting $\psi_{\tau} > 0$ with $\beta_{\tau} \approx 0$. A uniform response across sectors would indicate that we are capturing something other than input complementarity. **This is a test the existing literature cannot run, because it requires a sectoral elasticity rather than a complexity proxy.** The same proposition delivers a second restriction along a distinct margin: the response should be increasing in N_s , since exposure can only be uneven across inputs if there are several of them. We therefore interact exposure with N_s and predict a positive coefficient.

A further restriction concerns the composition of the response rather than its magnitude. As noted above, our model predicts that cost shocks and risk shocks operate through different channels and therefore produce different patterns of adjustment. We test this by estimating (2.18) for the 2018–19 tariff escalation, a cost shock with a China-specific signature, and comparing the estimated response along each margin. We predict that the tariff shock reduces the China share without raising within-input redundancy, while the pandemic raises redundancy and lowers Δ . This is a joint test of the model’s two channels, and because the tariff episode lies inside our pre-pandemic window it also serves as a placebo for the pandemic design.

Two conventional checks complete the exercise. Equation (2.18) includes seventy-two monthly leads, and we report the full set together with a test of their joint significance. We also test the belief-updating interpretation of the mechanism directly: if firms adjust because they revise their assessment of regional risk, those that reconfigured after 2020 should respond less to subsequent disruptions. We examine this using the 2021 Southeast Asian wave, interacting later exposure with the magnitude of the firm’s earlier adjustment.

3 A model of global sourcing with correlated risk

We now develop a quantitative model of firm-level sourcing that embeds the trade-off documented in Section 2. The model combines three ingredients. From Antràs et al. (2017) we take the structure of the firm’s sourcing problem: heterogeneous firms pay a fixed cost to establish the capability to source from each foreign country, and subsequently buy each input from the cheapest available supplier within the chosen set. From Lind and Ramondo (2023) we take a correlated multivariate Fréchet distribution for unit costs, which allows comparative advantage to be geographically clustered: countries in the same region draw similar costs, so that a geographically concentrated supply chain forgoes the gains from specialization. Finally, following Grossman et al. (2023), inputs are subject to exogenous risks of supply disturbance; we depart from their two-country formulation by giving those disturbances a *regional factor structure*, so that the correlation of delivery failures across sourcing locations is itself a function of geography.

The model’s central object is new to this literature. Because production requires many distinct and imperfectly substitutable input types, and because exploiting comparative advantage means assigning each input type to the region that produces it most cheaply, comparative advantage mechanically generates *uneven regional exposure across input types*. We show that input complementarity penalizes precisely this unevenness, and not geographic dispersion *per se*: two firms with identical sourcing-set size and identical average pairwise distance can face very different disruption risk, depending on whether each individual input is sourced from a single region or spread across several.

3.1 Environment and timing

There are $J + 1$ locations indexed $j \in \mathcal{L} \equiv \{0, 1, \dots, J\}$, where $j = 0$ denotes the home country. Locations are partitioned into K mutually exclusive and exhaustive geographic regions, $\mathcal{L} = \bigcup_{k=1}^K \mathcal{R}_k$, and we write $k(j)$ for the region containing location j . Wages w_j and iceberg trade costs τ_j from j to home are exogenous; we treat Mexico as a small open economy so that foreign wages and the demand shifter faced by Mexican firms are unaffected by Mexican sourcing decisions. **Appendix ?? discusses the general-equilibrium closure.**

Each firm belongs to a sector s , produces a single differentiated variety, and is characterized by a core productivity φ drawn from a Pareto distribution with shape κ and lower bound $\underline{\varphi}$, as in Melitz (2003). Timing within a period is as follows.

1. The firm observes φ and chooses its *sourcing set* $\mathcal{J} \subseteq \mathcal{L}$, with $0 \in \mathcal{J}$ by assumption, paying a fixed cost $w_0 f_j$ in home labor for each $j \in \mathcal{J} \setminus \{0\}$. The set is chosen before

any cost or delivery shocks are realized.

2. Unit-cost draws $\{z_j^m(\nu)\}$ are realized for every location j , input type m and variety ν . The firm identifies its cheapest supplier for each variety within $\mathcal{J}$, places its input orders, and posts a price $p(\nu)$.
3. Delivery shocks are realized. Suppliers that are disrupted fail to deliver; within the period the firm cannot re-source from an alternative location. The firm pays only for inputs that arrive, produces whatever its realized input bundle permits, and sells at the posted price p .

Two features of this timing deserve emphasis. First, the sourcing set is chosen *ex ante* and is therefore a portfolio: a location included in $\mathcal{J}$ delivers both a chance at a low cost draw and insurance against the failure of other members of the set. Second, prices are posted before disruptions are observed, so that a disruption results in unfilled demand rather than a price increase. This is what makes disruption risk a first-order determinant of expected profits rather than a **second-order Jensen effect**, and it is consistent with the short-run rigidities documented by Boehm et al. (2019) and Peter and Ruane (2026).

3.2 Demand at export markets

Firms face a constant-elasticity residual demand for their variety,

$$q(p) = Ap^{-\sigma}, \quad \sigma > 1. \quad (3.1)$$

where A is a demand shifter. In the quantification we interpret (3.1) as demand from the firm's export market, consistent with our use of firm-level exports as the outcome variable in Section 2; A then absorbs the destination price index and aggregate expenditure. Nothing below depends on this interpretation.

3.3 Input types and varieties

Production in sector s requires N_s distinct *input types*, indexed $m = 1, \dots, N_s$.^{*} They are combined with a sector-specific elasticity of substitution ζ_s ,

$$Q(\varphi) = \varphi \left[\sum_{m=1}^{N_s} a_m^{1/\zeta_s} X_m^{\frac{\zeta_s-1}{\zeta_s}} \right]^{\frac{\zeta_s}{\zeta_s-1}}, \quad a_m = \frac{1}{N_s}, \quad \sum_m a_m = 1. \quad (3.2)$$

^{*}For example, input types are the “essential inputs” needed for building a computer: the central processing unit, the display panel and the power supply.

We are interested in the empirically relevant case $\zeta_s < 1$, so that input types are gross complements; the Leontief benchmark is obtained as $\zeta_s \rightarrow 0$, and the Cobb-Douglas benchmark as $\zeta_s \rightarrow 1$. Estimates of ζ_s in the literature are uniformly below unity: Atalay (2017) and Gosselin (2025) report values well inside $(0, 1)$ for U.S. industries, and Boehm et al. (2019) report values close to zero for firms heavily reliant on imported inputs.

Each input type is itself a continuum of *varieties* $\nu \in [0, 1]$, aggregated with elasticity η ,

$$X_m = \left[\int_0^1 x_m(\nu)^{\frac{\eta-1}{\eta}} d\nu \right]^{\frac{\eta}{\eta-1}}, \quad \eta > 1. \quad (3.3)$$

Varieties within a type are gross substitutes; different suppliers of a capacitor are close substitutes for one another, whereas a capacitor is not a substitute for a display panel. This two-tier structure is what allows the model to separate the two distinct notions of complementarity that the existing literature conflates: substitutability *across suppliers of the same input*, estimated by Fujiy et al. (2024) and Barrot and Sauvagnat (2016), and substitutability *across input categories*, estimated by Atalay (2017) and Peter and Ruane (2026). The former maps to η , the latter to ζ_s .

Assumption 1 (Type-specific comparative advantage). *The Fréchet scale parameter T_j^m varies across input types m as well as across locations j .*

Assumption 1 is the formal counterpart of the revealed comparative advantage patterns documented in Section 2: Germany is relatively productive in machinery components, Vietnam in assembly. It is the reason different input types are optimally sourced from different regions, and hence the reason exposure to supply disruptions is uneven.

3.4 Sourcing and correlated comparative advantage

The cost of acquiring one unit of variety ν of input type m from location j is $\tau_j w_j / z_j^m(\nu)$, where $z_j^m(\nu)$ is a location-, type- and variety-specific productivity draw, interpreted as the unit-labor requirements to produce one unit of the variety. Following Lind and Ramondo (2023), we assume that the vector of draws $\mathbf{z}^m(\nu) = (z_0^m(\nu), \dots, z_J^m(\nu))$ follows a max-stable multivariate Fréchet distribution with a nested correlation structure,

$$\Pr(z_0^m \leq \bar{z}_0, \dots, z_J^m \leq \bar{z}_J) = \exp \left\{ - \sum_{k=1}^K \left[\sum_{j \in \mathcal{R}_k} (T_j^m \bar{z}_j^{-\theta})^{\frac{1}{1-\chi_k}} \right]^{1-\chi_k} \right\} \quad (3.4)$$

where $\theta > \eta - 1$ is the shape parameter governing the dispersion of draws and $\chi_k \in [0, 1)$ governs the correlation of draws within region k . The marginal distribution of each z_j^m is

Fréchet with scale T_j^m and shape θ for any $\{\chi_k\}$, so the correlation parameters are separately identified from the level of comparative advantage.

The parameter χ_k is the model's representation of Figure ?? . When $\chi_k = 0$ for all k , draws are independent and we recover Antràs et al. (2017) exactly. As $\chi_k \rightarrow 1$, locations within region k become perfectly correlated and hence redundant from a comparative advantage standpoint: adding a second country from the same region contributes nothing to the firm's cost. Intermediate values interpolate. Because our empirical evidence shows that revealed comparative advantage similarity declines with distance, we expect χ_k to be bounded well away from zero.

Define the *cost-adjusted capability* of location j in input type m ,

$$t_j^m \equiv T_j^m (\tau_j w_j)^{-\theta} \quad (3.5)$$

and the *sourcing capability* of set $\mathcal{J}$ in type m ,

$$\Theta_m(\mathcal{J}) \equiv \sum_{k=1}^K \left[\sum_{j \in \mathcal{J} \cap \mathcal{R}_k} (t_j^m)^{\frac{1}{1-\chi_k}} \right]^{1-\chi_k} \quad (3.6)$$

Lemma 1 (Prices and sourcing shares). *Under (3.4), the minimum cost of acquiring variety ν of type m within $\mathcal{J}$ is distributed $\Pr(c \leq x) = 1 - \exp\{-\Theta_m(\mathcal{J})x^\theta\}$, independently across varieties. Consequently:*

- (i) *The ideal price index of input type m is $P_m(\mathcal{J}) = \gamma \Theta_m(\mathcal{J})^{-1/\theta}$ with $\gamma \equiv \Gamma(\frac{\theta+1-\eta}{\theta})^{1/(1-\eta)}$*
- (ii) *The share of type- m expenditure allocated to location $j \in \mathcal{J} \cap \mathcal{R}_k$ is*

$$\pi_j^m(\mathcal{J}) = \frac{(t_j^m)^{\frac{1}{1-\chi_k}} \left[\sum_{i \in \mathcal{J} \cap \mathcal{R}_k} (t_i^m)^{\frac{1}{1-\chi_k}} \right]^{-\chi_k}}{\Theta_m(\mathcal{J})} \quad (3.7)$$

and the share allocated to region k is

$$\pi_k^m(\mathcal{J}) = \frac{\left[\sum_{j \in \mathcal{J} \cap \mathcal{R}_k} (t_j^m)^{\frac{1}{1-\chi_k}} \right]^{1-\chi_k}}{\Theta_m(\mathcal{J})} \quad (3.8)$$

- (iii) *The distribution of the price paid for a variety is independent of the identity of the supplying location. Hence the measure of varieties sourced from j equals the expenditure share π_j^m .*

Proof: See Appendix ?? for the appropriate derivations. Part (i) follows from the max-stability of (3.4) and homogeneity of degree one of the exponent, which give $\Pr(\min_j \tau_j w_j / z_j^m \geq x) = \exp\{-\Theta_m x^\theta\}$; the condition $\theta > \eta - 1$ ensures the relevant moment exists. Part (ii) is Euler's theorem applied to the exponent of (3.4). Part (iii) is the standard Eaton-Kortum independence property, which survives the introduction of correlation under max-stability. $\square$

Part (iii) of Lemma 1 is what makes the model tractable once delivery risk is introduced: it implies that the fraction of a type's input bundle lost when a set of locations is disrupted is simply the sum of their sourcing shares, with no need to keep track of *which* varieties were sourced where.

The firm's unit cost, given $\mathcal{J}$ and absent disruption, follows from (3.2) and Lemma 1(i):

$$c(\varphi, \mathcal{J}) = \frac{1}{\varphi} C(\mathcal{J}) = \frac{1}{\varphi} \left[\sum_{m=1}^{N_s} a_m P_m(\mathcal{J})^{1-\zeta_s} \right]^{\frac{1}{1-\zeta_s}} = \frac{\gamma}{\varphi} \left[\sum_m a_m \Theta_m(\mathcal{J})^{\frac{\zeta_s-1}{\theta}} \right]^{\frac{1}{1-\zeta_s}} \quad (3.9)$$

Note that ζ_s enters (3.9) only through the aggregation of type-level price indices. If comparative advantage were type-neutral ($T_j^m = T_j \forall m$), all P_m would coincide and ζ_s would drop out of the cost function entirely. Input complementarity therefore has no effect on costs in the absence of Assumption 1, and, as we show below, no effect on disruption either. The interaction between type-specific comparative advantage and input complementarity is the engine of the model.

Finally, define the *ex-ante expenditure share* of input type m ,

$$b_m(\mathcal{J}) \equiv \frac{a_m P_m(\mathcal{J})^{1-\zeta_s}}{\sum_{m'} a_{m'} P_{m'}(\mathcal{J})^{1-\zeta_s}}, \quad \sum_m b_m = 1 \quad (3.10)$$

3.5 Delivery risk with a regional factor structure

We now introduce disruption. Following Grossman et al. (2023), inputs are subject to exogenous risks of supply disturbance; our departure is to give those risks a factor structure that generates the geographic correlation pattern documented in Figure 21 and Figure ??.

Assumption 2 (Regional factor structure of delivery shocks). *Location j fails to deliver variety ν of type m if and only if at least one of three independent Bernoulli events occurs:*

$$\text{Failure}_j(\nu) \equiv \max \{ G, R_{k(j)}, \varepsilon_j(\nu) \} \quad (3.11)$$

where $G \sim \text{Bern}(g)$ is a global factor, $R_k \sim \text{Bern}(r_k)$ is a factor specific to region k , and $\varepsilon_j(\nu) \sim \text{Bern}(e_j)$ is idiosyncratic, drawn independently across locations and varieties. The

regional factors $(R_1, \dots, R_K)$ may be correlated with one another; the global factor and the idiosyncratic shocks are independent of everything else.

Assumption 2 is the model counterpart of the two-state example in Table ?? of Section ??, generalized to K regions. It parsimoniously reproduces the empirical facts of Section ??: two locations in the same region have correlated failures because they share R_k , whereas two locations in different regions share only G . Allowing the vector $(R_1, \dots, R_K)$ to be correlated permits the correlation of failures to decline smoothly with distance rather than stepping down discretely at region boundaries; in the quantification we implement this with a two-level structure in which $R_k = \max\{B_{b(k)}, \tilde{R}_k\}$ and $b(k)$ indexes broader macro-regions.

Two properties make (3.11) preferable to a multivariate normal specification for delivery shocks. First, a disruption is a discrete event: the shipment either arrives or it does not. A normal shock is unbounded in both directions and lacks this interpretation. Second, and decisively for computation, because $\varepsilon_j(\nu)$ is drawn independently across the continuum of varieties, idiosyncratic risk washes out by a law of large numbers within each input type. The firm's problem therefore depends on the realization of the aggregate factors only. We denote the aggregate state by

$$\mathbf{S} \equiv (G, R_1, \dots, R_K) \in \{0, 1\}^{K+1} \quad (3.12)$$

so that the state space has 2^{K+1} elements. With $K \approx 6$ to 8, expectations over $\mathbf{S}$ are computed exactly by enumeration.

Lemma 2 (Input losses). *Fix a sourcing set $\mathcal{J}$ and a state $\mathbf{S}$. The share of type- m input expenditure that fails to be delivered is*

$$\mu_m(\mathbf{S}; \mathcal{J}) = \begin{cases} 1 & \text{if } G = 1 \\ \sum_{k: R_k=1} \pi_k^m(\mathcal{J}) + \sum_{k: R_k=0} \sum_{j \in \mathcal{J} \cap \mathcal{R}_k} \pi_j^m(\mathcal{J}) e_j & \text{if } G = 0 \end{cases} \quad (3.13)$$

Proof: Immediate from Lemma 1(iii), which equates measure shares with expenditure shares, together with the law of large numbers applied to $\varepsilon_j(\nu)$ across ν . $\square$

Equation (3.13) makes the risk channel transparent. A firm that sources type m entirely from region k loses that input type completely whenever $R_k = 1$. A firm that splits type m across two regions loses only part of it. The regional *composition* of sourcing within each input type, not the number of countries or the average distance between them, is what determines exposure.

3.6 Production under supply disruptions

Because prices are posted and orders placed before shocks are realized, and because the firm cannot re-source within the period, a fraction $\mu_m(\mathbf{S})$ of its planned type- m purchases simply fails to arrive.

Lemma 3 (Within-type losses). *Let $\bar{X}_m$ denote the firm's planned type- m input bundle. If a set of varieties accounting for expenditure share μ_m fails to be delivered, the realized bundle is*

$$X_m(\mathbf{S}) = \bar{X}_m (1 - \mu_m(\mathbf{S}))^\psi, \quad \psi \equiv \frac{\eta}{\eta - 1} > 1. \quad (3.14)$$

Proof: Cost minimization implies $\bar{x}_m(\nu) = (p_m(\nu)/P_m)^{-\eta} \bar{X}_m$, so that the expenditure share of variety ν is $(p_m(\nu)/P_m)^{1-\eta}$. Setting $x_m(\nu) = 0$ on a set $\mathcal{S}$ with $\int_{\mathcal{S}} (p_m(\nu)/P_m)^{1-\eta} d\nu = \mu_m$ and substituting into (3.3),

$$X_m = \left[\int_{\nu \notin \mathcal{S}} \bar{x}_m(\nu)^{\frac{\eta-1}{\eta}} d\nu \right]^{\frac{\eta}{\eta-1}} = \bar{X}_m \left[\int_{\nu \notin \mathcal{S}} \left(\frac{p_m(\nu)}{P_m} \right)^{1-\eta} d\nu \right]^{\frac{\eta}{\eta-1}} = \bar{X}_m (1 - \mu_m)^\psi \quad \square$$

The exponent $\psi > 1$ reflects love of variety: losing ten percent of one's suppliers costs more than ten percent of the input bundle, because the surviving suppliers are imperfect substitutes. As $\eta \rightarrow \infty$, $\psi \rightarrow 1$ and losses are proportional.

Combining Lemma 3 with the outer aggregator (3.2) yields the object at the center of the model.

Definition 1 (Disruption index). The *disruption index* $\mathcal{D}(\mathbf{S}; \mathcal{J})$ is the ratio of realized to planned output,

$$\mathcal{D}(\mathbf{S}; \mathcal{J}) \equiv \left[\sum_{m=1}^{N_s} b_m(\mathcal{J}) (1 - \mu_m(\mathbf{S}; \mathcal{J}))^{\psi \frac{\zeta_s - 1}{\zeta_s}} \right]^{\frac{\zeta_s}{\zeta_s - 1}} \in [0, 1] \quad (3.15)$$

Definition 2 (Delivery index). Similarly define the *delivery index* $\Omega(\mathbf{S}; \mathcal{J})$, the share of planned input expenditure that the firm actually receives and pays for,

$$\Omega(\mathbf{S}; \mathcal{J}) \equiv \sum_{m=1}^{N_s} b_m(\mathcal{J}) (1 - \mu_m(\mathbf{S}; \mathcal{J})) \quad (3.16)$$

The wedge between Ω and $\mathcal{D}$ is economically meaningful: it is *waste*, inputs that arrive but cannot be used because complementary inputs did not. A computer manufacturer that receives its shipment of keyboards but not its shipment of processors pays for the keyboards and produces nothing.

The disruption index has three limiting cases, each of which recovers a familiar benchmark:

$$\zeta_s \rightarrow 0 : \quad \mathcal{D} = \min_m (1 - \mu_m)^\psi \quad (\text{Leontief; weakest link}) \quad (3.17)$$

$$\zeta_s \rightarrow 1 : \quad \mathcal{D} = \prod_m (1 - \mu_m)^{\psi b_m} \quad (\text{Cobb-Douglas; geometric mean}) \quad (3.18)$$

$$\eta \rightarrow \infty, \mu_m \equiv \mu : \quad \mathcal{D} = 1 - \mu \quad (\text{proportional losses}) \quad (3.19)$$

3.7 Input complementarity and uneven exposure

Equation (3.15) is a weighted power mean of order $(\zeta_s - 1)/\zeta_s$ in the variables $(1 - \mu_m)^\psi$. Two properties of power means then deliver the model's main qualitative results.

Proposition 1 (Neutrality under symmetric exposure). *Suppose input losses are equalized across types, $\mu_m(\mathbf{S}) = \mu(\mathbf{S})$ for all m and all $\mathbf{S}$. Then*

$$\mathcal{D}(\mathbf{S}; \mathcal{J}) = (1 - \mu(\mathbf{S}))^\psi \quad (3.20)$$

which is independent of ζ_s . Input complementarity has no effect on disruption.

Proof: Substituting $\mu_m = \mu$ into (3.15), the exponents $\psi(\zeta_s - 1)/\zeta_s$ and $\zeta_s/(\zeta_s - 1)$ cancel and the weights b_m sum to one. $\square$

Proposition 1 is worth dwelling on, because it corrects an intuition that is widespread in the policy discussion and implicit in our own Section ???. Complementarity does *not* amplify disruptions in general. If a shock hits all of a firm's inputs proportionally, a Leontief producer and a Cobb-Douglas producer suffer identically. What complementarity punishes is the *unevenness* of the shock across input types.

Proposition 2 (Complementarity and exposure dispersion). *Fix the vector of losses $\boldsymbol{\mu} = (\mu_1, \dots, \mu_{N_s})$ and the weights $\mathbf{b}$. Then:*

- (i) $\mathcal{D}$ is weakly increasing in ζ_s , strictly so whenever the μ_m are not all equal.
- (ii) $\mathcal{D}$ is **Schur-concave** in $((1 - \mu_1)^\psi, \dots, (1 - \mu_{N_s})^\psi)$ under equal weights: a mean-preserving spread in losses across input types strictly reduces output.
- (iii) The magnitude of the reduction in (ii) is strictly decreasing in ζ_s .

Proof: Write $z_m \equiv (1 - \mu_m)^\psi$ and $r \equiv (\zeta_s - 1)/\zeta_s$, so that $\mathcal{D} = M_r(\mathbf{z}; \mathbf{b})$, the weighted power mean of order r . For $\zeta_s \in (0, 1)$ we have $r < 0$. Part (i) follows because r is increasing in ζ_s

on $(0, 1)$ and power means are increasing in their order, with equality only if all z_m coincide. Part (ii) follows because M_r is symmetric and concave for $r \leq 1$, hence Schur-concave. Part (iii) follows from the fact that the curvature of $x \mapsto x^r$ increases as r falls. $\square$

The final step connects this to comparative advantage.

Proposition 3 (Comparative advantage generates uneven exposure). *Suppose Assumption 1 holds and let λ index the degree of regional specialization of comparative advantage, in the sense that t_j^m is a mean-preserving spread across m of a common t_j that increases with λ . Then the cross-type dispersion of regional exposures, $\{\pi_k^m\}_m$, is increasing in λ for any fixed $\mathcal{J}$, and hence, by Proposition 2, $\mathbb{E}[\mathcal{D}]$ is decreasing in λ .*

Taken together, Propositions 1–3 articulate the paper’s mechanism precisely: exploiting comparative advantage means assigning each input type to the region that produces it most cheaply. Doing so mechanically concentrates each input type’s exposure on a single region, making losses uneven across types. Uneven losses are exactly what input complementarity punishes. The trade-off between efficiency and risk therefore operates through the regional composition of sourcing within input types, and is stronger in sectors with lower ζ_s and larger N_s .

Definition 3 (Exposure dispersion). Let $\bar{\pi}_k(\mathcal{J}) \equiv \sum_m b_m \pi_k^m(\mathcal{J})$ denote firm-level regional sourcing shares. The firm’s *exposure dispersion* is

$$\Delta(\mathcal{J}) \equiv \sum_{m=1}^{N_s} b_m \sum_{k=1}^K (\pi_k^m(\mathcal{J}) - \bar{\pi}_k(\mathcal{J}))^2 \quad (3.21)$$

Δ is zero when every input type is sourced from the same regional mix, and is maximized when each input type is sourced entirely from a single, distinct region. Crucially, Δ is orthogonal by construction to the firm-level regional shares $\bar{\pi}_k$: two firms with identical aggregate geography can differ arbitrarily in Δ . It is computable directly from customs data as the between-product dispersion of import origin shares, and Section 4.1 shows that it, rather than average pairwise distance, is the model-implied sufficient statistic for disruption risk.

3.8 Pricing and expected profits

The firm posts p before observing $\mathbf{S}$, sells $\min\{Q(\mathbf{S}), Ap^{-\sigma}\}$, and pays only for delivered inputs. Planning for the undisrupted quantity $\bar{Q} = Ap^{-\sigma}$, expected profits gross of fixed costs are

$$\mathbb{E}[\pi^v] = p \bar{Q} \mathbb{E}[\mathcal{D}] - c(\varphi, \mathcal{J}) \bar{Q} \mathbb{E}[\Omega] \quad (3.22)$$

Proposition 4 (Optimal price and expected profits). *The profit-maximizing price is a constant markup over a risk-adjusted marginal cost,*

$$p^*(\varphi, \mathcal{J}) = \frac{\sigma}{\sigma - 1} \tilde{c}(\varphi, \mathcal{J}), \quad \tilde{c}(\varphi, \mathcal{J}) \equiv c(\varphi, \mathcal{J}) \frac{\mathbb{E}[\Omega(\mathcal{J})]}{\mathbb{E}[\mathcal{D}(\mathcal{J})]} \geq c(\varphi, \mathcal{J}). \quad (3.23)$$

and expected variable profits are

$$\mathbb{E}[\pi^v(\varphi, \mathcal{J})] = \frac{1}{\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{1-\sigma} A \varphi^{\sigma-1} C(\mathcal{J})^{1-\sigma} \Lambda(\mathcal{J}) \quad (3.24)$$

where the resilience index is

$$\Lambda(\mathcal{J}) \equiv \mathbb{E}[\mathcal{D}(\mathcal{J})]^\sigma \mathbb{E}[\Omega(\mathcal{J})]^{1-\sigma} \in (0, 1] \quad (3.25)$$

Proof: Substituting $\bar{Q} = A p^{-\sigma}$ into (3.22) and taking the first-order condition in p yields (3.23). Substituting back and collecting terms gives (3.24). That $\Lambda \leq 1$ follows from $\mathcal{D} \leq \Omega^\psi \leq \Omega \leq 1$ and $\sigma > 1$. $\square$

Equation (3.24) is the model's central expression and it states the trade-off compactly. Expected profits are the product of two terms that move in opposite directions as the supply chain becomes more geographically dispersed: $C(\mathcal{J})^{1-\sigma}$ is *increasing* in dispersion. Because $\chi_k > 0$, adding a country from a new region raises Θ_m by more than adding a country from a region already represented, lowering C . This is the comparative advantage force of Figure 20, and its strength is governed by θ and $\{\chi_k\}$. $\Lambda(\mathcal{J})$ is *decreasing* in dispersion whenever regional shocks are imperfectly correlated. Spreading across regions raises the probability that *some* input type is disrupted, and by Proposition 2 the cost of such disruptions rises with complementarity. This is the risk force of Figure 21, and its strength is governed by $\{r_k\}$, g , ζ_s and N_s . Both terms are available in closed form given $\mathcal{J}$, which is what makes the model quantifiable.

3.9 The input sourcing decision

The firm's sourcing problem is the following

$$\mathcal{J}^*(\varphi) = \arg \max_{\mathcal{J} \subseteq \mathcal{L}} \left\{ \frac{1}{\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{1-\sigma} A \varphi^{\sigma-1} C(\mathcal{J})^{1-\sigma} \Lambda(\mathcal{J}) - w_0 \sum_{j \in \mathcal{J} \setminus \{0\}} f_j \right\} \quad (3.26)$$

A firm produces if and only if the maximized value is non-negative, which defines a productivity cutoff φ^* as in Melitz (2003).

Problem (3.26) is a combinatorial discrete choice problem over 2^J sets. In Antràs et al. (2017) the corresponding objective is supermodular under an empirically relevant parameter restriction, which permits the lattice algorithm of Jia (2008). Our objective is *not* supermodular in general, for two distinct reasons:

1. Correlated cost draws make locations within a region partial substitutes. Adding a second country from region k raises Θ_m by less when a country from k is already in $\mathcal{J}$, and the effect is stronger the larger is χ_k .
2. Correlated delivery risk makes the marginal value of a location depend on the regional composition of the incumbent set in a sign-indeterminate way. Adding a country from a region already represented reduces Λ by little; adding a country from an unrepresented region may reduce it substantially, or may *raise* it if doing so reduces exposure dispersion Δ .

We therefore solve (3.26) using the methods of Arkolakis et al. (forthcoming), whose single-crossing-differences condition is weaker than supermodularity and admits both complementarities and substitutabilities among locations. **Appendix ?? verifies that the condition holds at our estimated parameters and reports the algorithm in full.** As a robustness check we also solve a restricted two-stage version in which the firm first allocates expenditure shares across the K regions — a convex problem with an interior solution — and then chooses countries within each region, where the objective is supermodular and Jia (2008) applies directly. The two solutions coincide closely in our parameterization.

3.10 Discussion of key assumptions

Five assumptions do substantive work in the model, and we set out the justification for each and the sense in which our results depend on it.

The central rigidity is that disrupted inputs cannot be replaced within the period. This is supported by the short-run elasticity estimates in the literature: Boehm et al. (2019) find that output falls roughly one-for-one with imported input declines for firms heavily reliant on Japanese inputs; Fujiy et al. (2024) estimate an elasticity of substitution across suppliers of the same product of about 0.55; and Peter and Ruane (2026) document that elasticities estimated from transitory shocks lie below one even where long-run elasticities are well above one. In **Appendix ?? we relax the assumption by allowing a fraction $\lambda \in [0, 1]$ of disrupted varieties to be re-sourced at a proportional penalty, and show that the qualitative**

results survive for λ below a threshold. A closely related timing assumption is that prices are posted before shocks are realized, so that disruption generates unfilled orders rather than price increases and risk affects the level of expected profits at first order. Under ex-post pricing, risk would matter only through the concavity of $\mathcal{D}$, a second-order effect that we find is an order of magnitude too small to rationalize the export volatility differences documented in Figure ??.

It is worth being explicit about what these two assumptions buy, because it is what allows us to dispense with a third. Firms in our model are risk neutral. The mean-variance representation that motivates the paper is a *result* of the interaction between complementarity and correlated shocks, not an assumption about preferences. This distinguishes our approach from portfolio formulations that impose risk aversion directly, and it means that our quantitative results do not hinge on a free risk-aversion parameter.

The remaining two assumptions concern measurement and scope. The number of input types N_s enters both the cost function, where it is absorbed into the productivity distribution, and the disruption index, where it determines how finely exposure can vary across types; it is therefore not separately identified from cost data alone. We measure it directly, as the effective number of inputs in sector s implied by the Mexican input-output matrix, and validate it against the effective number of distinct HS6 products imported by firms in that sector. Delivery risk, finally, is exogenous: firms cannot invest in resilience beyond their choice of $\mathcal{J}$, nor hold inventories. Grossman et al. (2023) show that in such an environment private diversification decisions need not be socially optimal, so our counterfactuals should be read as positive rather than normative.

4 Quantifying the model

In here we should describe the entire quantification of the model.

4.1 Parameterization and identification

Table 2 summarizes the parameters. We set externally those for which credible estimates exist and estimate the remainder by simulated method of moments. Identification of the estimated parameters proceeds as follows.

- θ is identified by the elasticity of the firm’s export volume with respect to the size and geographic composition of its sourcing set, i.e. the slope in Figure ?. A larger θ means less dispersed cost draws and hence smaller gains from adding locations.

Parameter	Interpretation	Source	Value
<i>Set externally</i>			
σ	Demand elasticity	Literature	4.0
η	Substitution across suppliers	Fujiy et al. (2024)	0.55 [†]
ζ_s	Substitution across input types	Gosselin (2025)	sector-specific
N_s	Effective number of input types	INEGI input-output matrix	sector-specific
T_j^m	Comparative advantage	CEPII	country $\times$ type
κ	Pareto shape of φ	Literature	—
<i>Estimated by SMM</i>			
θ	Fréchet shape	—	—
$\{\chi_k\}$	Within-region cost correlation	—	—
g	Global disruption probability	—	—
$\{r_k\}$	Regional disruption probs.	—	—
$\{e_j\}$	Idiosyncratic disruption probs.	—	—
$\{f_j\}$	Fixed sourcing costs	—	—

[†] Because the model requires $\eta > 1$ for the inner aggregator to be well defined, we treat the Fujiy et al. (2024) estimate as a lower bound and report results across a range; see Appendix ??.

Table 2: Parametrization of the model

- $\{\chi_k\}$ are identified by the *regional* composition of firms’ import portfolios. High χ_k makes within-region countries redundant, so firms spread across regions; the observed regional-share distribution therefore pins down the χ_k . This is reinforced by the country-level relationship between distance and RCA similarity in Figure 20.
- $\{f_j\}$ are identified by the distribution of the number of sourcing countries and by the extensive margin of country participation across the productivity distribution, exactly as in Antràs et al. (2017).
- $\{e_j\}$ are identified by the number of origin countries per HS6 product. Idiosyncratic risk is the only force in the model that makes a firm hold multiple sources of the *same* input, since regional and global shocks cannot be diversified within a region.
- g and $\{r_k\}$ are identified by the coefficient of variation of firm exports and its gradient in exposure dispersion Δ . The global factor shifts the level of volatility common to all firms; the regional factors generate volatility that varies with Δ . It is the interaction of Δ with the sectoral parameters (ζ_s, N_s) that separates $\{r_k\}$ from g .

We estimate on the pre-pandemic sample, 2014–2019, and reserve 2020 as an out-of-sample test. Specifically, we feed the observed 2020 realization of the **Oxford Stringency Index** into the model as an increase in g and $\{r_k\}$, calibrated to match the observed severity

and cross-regional pattern of lockdowns, and ask whether the model reproduces the collapse in chain size, efficiency and dispersion documented in Figures 23 and 24, and the differential export decline by risk exposure documented in Table ???. None of these moments is targeted in estimation.

4.2 A testable implication

Proposition 2 and Definition 3 generate a prediction that distinguishes this model from the reduced-form trade-off in Section ??? and that we take to the data in Section ???. Consider two firms in the same sector, with the same number of sourcing countries, the same average pairwise distance among them, and the same aggregate regional shares $\bar{\pi}_k$. Firm A sources each of its inputs from a single region, with different inputs assigned to different regions. Firm B sources each of its inputs from the same mix of regions. Then $\Delta_A > \Delta_B = 0$, and the model predicts:

- (i) Firm A has lower expected output and higher export volatility than firm B
- (ii) The gap is increasing in N_s and decreasing in ζ_s
- (iii) The gap is absent when $\zeta_s = 1$, regardless of geography

Prediction (iii) is a sharp falsification test: it says that the entire efficiency-risk trade-off should vanish in sectors whose inputs are Cobb-Douglas substitutes, even for firms with maximally dispersed supply chains. Existing measures of supply chain risk, which are built from average pairwise distances or correlations, cannot generate this prediction because they do not distinguish exposure dispersion from geographic dispersion.

5 Conclusions

This paper makes several novel contributions to the literature on global value chains (GVCs) by proposing a mean-variance trade-off in GVC formation using geographical dispersion as the key variable. We argue that firms face a trade-off between exploiting comparative advantage by sourcing inputs from distant locations and minimizing the risk of supply chain disruptions due to asynchronous business cycles. Our empirical analysis, using both country-level data and detailed firm-level customs data from Mexico, provides evidence supporting this trade-off. We also present new empirical facts regarding recent trends in GVCs, highlighting how, before COVID-19, firms were increasingly sourcing inputs from a larger number of locations and from locations that were more geographically dispersed,

thereby enhancing their comparative advantage but also exposing themselves to higher risks of disruption. These trends reversed sharply in 2020 in response to the trade collapse driven by global lockdowns and restrictions. For future research, a more refined proxy for sectoral heterogeneity and complementarity is essential. The current use of the HS code, while useful, may not fully capture the degree of complementarity in input requirements and production complexities. We also propose numerically solving the model to explore the effects of varying key parameters that drive the tradeoffs in GVC formation. By employing the method of moments, we can estimate these parameters more accurately and analyze their impact on the mean-variance tradeoff. This approach would allow us to simulate different scenarios and better understand the conditions under which firms optimize their GVC strategies, providing deeper insights into the policy implications and potential strategies to mitigate supply chain risks.

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